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MeCP2 regulates telencephalic development in human cerebral organoids

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AUTHOR CONTRIBUTIONS

M.F.T. designed the study, performed all experiments, analyzed and interpreted the data, and wrote the manuscript. R.M.-B. assisted with cloning, bacterial work, and antibody titration for cell hashing. S.M.S.-S. and A.R.M.D. contributed to the puromycin selection experiments and cell cryopreservation. R.B. determined optimal AAV dosing and helped with AAV and LV infection experiments. R.M.-B. and S.M.S.-S. contributed equally to this study. A.R.M. supervised the work and secured funding.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Dr. Alysson R. Muotri (muotri@health.ucsd.edu).

Materials availability

This study did not generate new, unique reagents. The RTT iPSC lines used in this study can be obtained via the Rett Syndrome Research Trust (RSRT).

Data and code availability

- The bulk RNA-seq and scRNA-seq data generated in this article have been deposited in NCBI's Gene Expression Omnibus (GEO: GSE297594).
- Additional information required to reanalyze the data reported in this manuscript will be shared by the lead contact upon reasonable request.
- This article does not report any original code.

DECLARATION OF INTERESTS

A.R.M. is a co-founder and has an equity interest in TISMOO, a company dedicated to genetic analysis and human brain organogenesis, focusing on therapeutic applications customized for autism spectrum disorders and other neurological conditions. A.R.M. is also an inventor on several patents using brain/cerebral organoids. The terms of this arrangement have been reviewed and approved by the University of California, San Diego, in accordance with its conflict-of-interest policies.

SUPPLEMENTAL INFORMATION

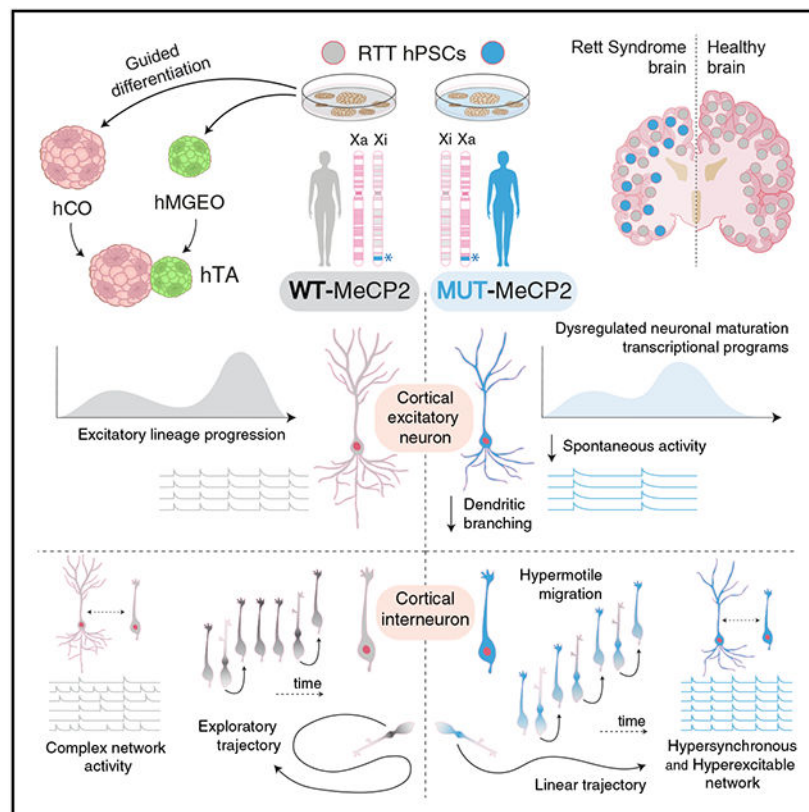
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SUMMARY

Loss-of-function variants in the X-linked gene *MeCP2* cause a severe form of syndromic autism spectrum disorder known as Rett syndrome (RTT). Although traditionally considered a postnatal disorder, increasing evidence suggests that *MeCP2* plays a crucial role during prenatal brain development. Here, we used human pluripotent stem cell-derived cerebral organoids and human telencephalic assembloids to model early cortical development in the context of *MeCP2* deficiency. Loss of *MeCP2* led to widespread dysregulation of transcriptional programs associated with cortical excitatory neuron development, accompanied by delayed morphological and functional maturation, despite preserved neuroepithelial architecture, dorsal telencephalic identity, and laminar organization. *MeCP2* deficiency also led to an overproduction of cortical interneurons (cINs), with these cINs exhibiting hypermotile tangential migration dynamics and contributing to persistent hypersynchronous network activity in assembloids. These findings highlight the critical role of *MeCP2* in early telencephalic neurodevelopment and underscore the prenatal origin of RTT-related dysfunction.

Graphical Abstract



In brief

Using cerebral organoids and assembloids, Tenreiro et al. describe that *MeCP2* deficiency has profound, lineage-specific effects during telencephalic development. These findings provide insights into the origins of Rett syndrome-associated neurological dysfunction.

INTRODUCTION

Methyl-CpG-binding protein 2 (MeCP2) is a broadly expressed epigenetic regulator known for its high-affinity binding to methylated DNA across the genome,¹⁻³ mediating both transcriptional repression^{4,5} and activation.⁶ Beyond transcriptional regulation, MeCP2 contributes to higher-order chromatin architecture,^{7,8} microRNA processing,⁹ and alternative splicing.¹⁰ Loss-of-function variants in *MeCP2* cause a severe and rare form of syndromic autism spectrum disorder (ASD), known as Rett syndrome (RTT; OMIM #312750),^{11,12} that predominantly affects females due to the gene's X-linked nature.^{13,14} Affected girls develop normally before 6–18 months of age, followed by regression with loss of motor and language skills, intellectual disability, and stereotypic behaviors.¹⁵ MeCP2 is highly enriched in the brain,^{16,17} with protein levels rising postnatally and remaining essential throughout life, as deletion in juvenile or adult mice reproduces RTT-like phenotypes.¹⁸⁻²¹

Although RTT symptoms arise postnatally, emerging evidence suggests prenatal MeCP2 functions. Subtle behavioral abnormalities can be detected in early infancy,²²⁻²⁴ and hemizygous male carriers show severe prenatal phenotypes, from neonatal encephalopathy²⁵ to embryonic lethality in non-human primates.²⁶ In mice, MeCP2 is expressed in the embryonic cortex,^{16,27} and its loss biases progenitors toward proliferation over neuronal differentiation.²⁷ Likewise, neurons derived from patient-derived RTT human pluripotent stem cells (hPSCs), which mimic fetal stages of development, show reduced dendritic complexity,^{28,29} supporting an early developmental role for MeCP2.

Notably, signs of altered neural activity in patients with RTT precede postnatal network refinement, indicating early circuit-level dysfunction. Abnormal network synchrony and hyperexcitability have emerged as converging features in several ASD mouse models—not only in *MeCP2*-deficient models of RTT^{30,31} but also in those harboring loss-of-function mutations in *FMR1*, *SHANK3*, and *UBE3A*.³² Although RTT pathology spans multiple neuronal subtypes and brain regions,^{30,33-39} γ -aminobutyric acid (GABA)ergic signaling appears particularly vulnerable, and deficits in inhibitory circuitry may underlie distinct features of the disorder, including stereotypic behaviors.³⁰ Inhibitory circuit maturation begins prenatally, and early oscillatory activity, such as delta brushes, is already detectable in the human fetal brain.⁴⁰ Disruptions in interneuron intrinsic properties or the modulatory role of GABA, including a delayed chloride reversal switch and prolonged depolarizing action, can have lasting consequences on circuit plasticity,³² potentially contributing to the epileptiform activity characteristic of RTT and other ASDs.⁴¹

Directly investigating the unfolding of these developmental processes in humans is inherently limited, as it would require access to and manipulation of fetal brain tissue. Cerebral organoids provide a powerful alternative for studying *MeCP2* function, recapitulating telencephalic developmental programs that are poorly represented in animal models⁴² and not captured in 2D neural cultures. More sophisticated 3D model systems,

such as assembloids—created by fusing regionally patterned organoids, including dorsal and ventral forebrain domains⁴³⁻⁴⁵—further enable exploration of higher-order telencephalic circuits that depend on inhibitory modulation. The latter approach has uncovered roles for several neurodevelopmental disorder-associated genes in early cortical interneuron (cIN) developmental processes, including their production and tangential migration.⁴⁶ Continued improvements in organoid technology have addressed key limitations in cell-type identity, patterning fidelity, and maturation,^{47,48} emphasizing the need for high-quality samples to accurately model human cortical development.⁴⁹ Such variability likely underlies discrepancies among prior RTT organoid studies, with some reporting altered excitatory neuron abundance,⁵⁰ whereas others found no changes in differentiation.⁵¹

Here, we report that *MeCP2* is expressed in the prenatal telencephalon and that loss-of-function variants disrupt developmental programs in both excitatory and inhibitory lineages. Using single-cell transcriptomics and functional analyses of *MeCP2*-deficient organoids and assembloids, we show that *MeCP2* regulates transcriptional programs of cortical excitatory neuron maturation and is required to restrain hypermotile cIN tangential migration and prevent hypersynchronicity and hyperexcitability in emerging cortical circuits. Our findings underscore that RTT is not solely a postnatal neurodevelopmental disorder, as *MeCP2* deficiency perturbs critical processes already underway in the perinatal period.

RESULTS

MeCP2 is expressed during cerebral cortex development

MeCP2 is best known for its role in postnatal brain maturation—consistent with RTT symptoms emerging in patients after the first year of life and in *Mecp2*-null mice around 4–5 weeks of age⁵²—yet it remains unresolved as to what extent prenatal dysfunction contributes to the disorder. Notably, conditional deletion of *Mecp2* in postnatal neurons recapitulates the full null phenotype, albeit with delayed onset, suggesting a postnatal requirement.⁵³ However, whether MeCP2 is required earlier, during cortical development, is still debatable, with hPSC-based studies showing inconsistencies in the phenotypes observed.^{50,51,54} To investigate this, we first analyzed bulk transcriptomic data from the BrainSpan Atlas,⁵⁵ which revealed *MeCP2* expression across all stages of human brain development, with a steady postnatal increase and already substantial levels during the second trimester—coinciding with peak cortical neurogenesis^{56,57} (Figure S1A). To resolve cell type-specific patterns, we examined single-cell RNA-sequencing data from human fetal neocortex (gestational week; GW17–18),⁵⁸ revealing broad *MeCP2* expression across both progenitor and neuronal populations, including radial glia as well as infragranular and supragranular neurons (Figure S1B).

Although *MeCP2* is broadly expressed, its protein levels vary across tissues, making transcriptomic data an imperfect proxy for functional expression.¹⁶ In mice, MeCP2 protein accumulation follows the “inside-out” logic of corticogenesis, with initial enrichment in Cajal-Retzius cells and infragranular neurons (layers V–VI), followed by supragranular neurons (layers II–IV).¹⁶ Human fetal tissue studies are limited by sample scarcity and potential protein degradation in *postmortem* specimens. To overcome this, we used human cerebral organoids (hCOs), which recapitulate the spatiotemporal dynamics of neocortical

development^{59,60} and allow monitoring of neurodevelopmental disease-associated gene expression.⁶¹ At 8 weeks, hCOs exhibited well-developed progenitor and cortical plate (CP)-like zones (Figure S1C), with MeCP2 protein broadly present but enriched in post-mitotic neurons, supporting a developmentally regulated role that intensifies with neuronal differentiation (Figure S1D).

MeCP2 deficiency does not impair the differentiation of dorsal telencephalic neurons

To understand how *MeCP2* deficiency influences dorsal telencephalic development, we reprogrammed blood lymphocytes from patients with classical features of RTT with known mutations in *MeCP2* (Figure S2A)—representing three out of the eight recurrent “hotspot” variants found in 60% of diagnosed cases of RTT^{62,63}—into human induced pluripotent stem cells (hiPSCs) that co-expressed the self-renewal pluripotency markers OCT4 and NANOG (Figure S2B). Because *MeCP2* is an X-linked gene, females with RTT are somatic mosaics, with cells either expressing or lacking a functional copy of this gene. During human somatic cell reprogramming, the inactive X chromosome (Xi) generally retains its silenced state,⁶⁴ allowing isolation of clonal hiPSCs expressing either the wild-type (WT) or mutant (MUT) allele exclusively (Figure 1A). As X chromosome inactivation (XCI) skewing can influence phenotypical variability seen in females with RTT, and even result in asymptomatic carriers,^{65,66} only clones with confirmed monoallelic expression of either WT or MUT *MeCP2* were used, verified via Sanger sequencing (Figure 1B; Figure S2C). Consistent with XCI maintenance, *Xist* RNA foci were detected in each clone by RNA-FISH (Figure 1C; Figure S2D). Female hPSC lines suffer from erosion of the Xi with prolonged *in vitro* culture,⁶⁷ and so we closely monitored *MeCP2* monoallelic status throughout the course of experiments (Figure S2E). To study *MeCP2* loss in an isogenic context, a homozygous *MeCP2* deletion was also introduced into a commonly used human embryonic stem cell (hESC) line (Figure 1A; Figure S2F), as previously described.⁶⁸

In the embryonic mouse brain, *MeCP2* deficiency has revealed an enrichment of cell proliferation over late post-mitotic neuronal identities²⁷ and a shift from neurogenesis to astrogliogenesis.^{54,69} Given that *MeCP2* is expressed during human corticogenesis, we sought to explore its role in dorsal telencephalic development by generating hCOs from our WT- and MUT-MeCP2 hPSCs (Figure 1D). By week 3.5 of differentiation, WT and MUT hCOs were morphologically indistinguishable and similar in size (Figure S3A), expressing canonical neural progenitor markers, such as SOX2 and Nestin, as well as forebrain- and dorsal telencephalon-associated transcription factors, namely FOXG1, LHX2, and PAX6 (Figure 1E; Figures S3B and S3C). The neuroepithelium initially displayed apical-out polarity but reorganized into rosette-like structures with apical-in polarity resembling the ventricular zone (VZ) at the luminal side of the dorsal telencephalic vesicle (Figures 1E and 1F; Figure S3D). The VZ serves as the primary germinal zone during early stages of neurogenesis and is populated by apical radial glia (aRG), which extend basal fibers to the pial surface, providing a cytoarchitectonic scaffold to guide newborn immature neurons toward their final destination in the developing CP.^{70,71} To explore MeCP2's role during the formation of the CP, we started by analyzing the morphology of VZs between weeks 5 and 6 of differentiation. Both WT and MUT hCOs formed well-organized pseudostratified VZs, but MUT lines showed modestly larger and thicker VZs (Figure 1G), most evident in lines

2042 and 2047 (Figure S3E). Because MeCP2 recruits the NCOR-SMRT corepressor to silence gene expression^{4,5} but can also promote transcription via interaction with CREB1,⁶ we performed bulk RNA sequencing on week 5–6 organoids from patient 2042 to investigate transcriptional changes underlying VZ expansion. Differentially expressed genes (DEGs) upregulated in MUT hCOs were associated with neural progenitor self-renewal (*HEY2*, *NOTCH2*,⁷² and *TP73*⁷³), extracellular matrix remodeling (*COL1A1* and *LAMA3*), and WNT signaling activation (*RSPO1*⁷⁴ and *WNT3A*⁷⁵; Figure S3F), which are known to promote VZ expansion in the dorsal telencephalon. Gene Ontology (GO) analysis supported this, revealing enrichment for processes involved in epithelial morphogenesis (Figure S3G). Conversely, WT hCOs upregulated neuronal differentiation genes (Figure S3F), while early cortical laminar markers were unchanged (Figure S3H).

To evaluate whether the slight VZ enlargement observed in MUT hCOs affects CP formation, we extended our analysis of organoid differentiation up to week 17 (~4 months). Neuronal subtype identity and laminar allocation within the CP are governed by the coordinated and temporally regulated expression of multiple transcription factors, including TBR1⁷⁶ (layer VI), FEZF2⁷⁷ and BCL11B/CTIP2⁷⁸ (layer V), ROR β ⁷⁹ (layer IV), and BRN2,⁸⁰ SATB2,⁸¹ and CUX1/2⁸² (layers II–III). Differentiation progressed comparably across genotypes. By weeks 5–6, both WT and MUT organoids displayed TBR1⁺ neurons encircling the VZs, consistent with early stages of CP genesis (pre-plate formation; Figure 1H; Figure S4A). At week 8, a clear CP domain expressing the infragranular marker BCL11B surrounded a SVZ-like zone enriched in EOMES/TBR2⁺ intermediate progenitors (IPs; Figure 1I; Figure S4B). By week 17, both genotypes showed robust SATB2 expression in the CP, marking supragranular callosal projection neurons (Figure 1J; Figure S4C). Throughout differentiation, MeCP2 immunoreactivity was observed in progenitor domains (SOX2⁺ VZs and EOMES⁺ IPs) and more strongly in the CP (Figures S4D–S4F). To rule out lamination defects, we quantified the distribution of BCL11B⁺ and SATB2⁺ neurons in the CP of patient 2042. Both WT and MUT hCOs exhibited the canonical inside-out layering pattern, with SATB2⁺ neurons in superficial and BCL11B⁺ neurons in deep layers—reflective of their respective birthdates—and no significant differences in their spatial distribution between genotypes (Figure S4G). Together, these data indicate that *MeCP2* mutations do not overtly impair corticogenesis in hCOs, despite an early enrichment of progenitor over neuronal identities.

Neocortical cell-fate acquisition is unaffected by MeCP2 deficiency

Given MeCP2's broad genomic binding (~11% of the genome³) and cell type-specific regulatory functions,⁸³ subtle transcriptional alterations might arise along the excitatory lineage despite a histologically normal corticogenesis. To investigate this possibility, we performed single-cell RNA sequencing on week 17 WT and MUT hCOs to examine the cell type-specific effects of *MeCP2* deficiency on neocortical development. Unbiased clustering of 55,478 cells from nine individual organoids, derived from two patients with distinct genetic backgrounds (patients 2036 and 2042), revealed eight clusters of telencephalic populations, namely aRG, outer radial glia (oRG), IPs, cINs with caudal ganglionic eminence (CGE) identity, and three subtypes of cortical excitatory neurons, distinguished by enrichment of infragranular or supragranular markers (or lacking a clear specification

in the case of immature neurons; Figure 1K; Figures S5A and S5B). Cluster identities, assigned using established markers characteristic of mid-gestation cell populations (Figure 1L; Figure S5C), showed strong concordance with *in vivo* fetal transcriptomic signature⁵⁸ (Figure 1M; Figure S5D) and significant spatial transcriptomic similarity to the E15.5 mouse telencephalon (Figure S5E), confirming that hCOs faithfully model dorsal telencephalic development.

We assessed cluster composition between genotypes within each patient group and did not observe any consistent differences between WT and MUT hCOs (Figure 1N; Figures S5F and S5G). While the proportion of cells per cluster was largely reproducible across individual organoids, the observed variability—such as that observed between 2036-WT and 2042-WT—may reflect inherent inter-individual heterogeneity typical of endogenous brain tissue.⁸⁴ Nonetheless, *MeCP2* was expressed throughout all developmental stages in our dataset (Figure 1O), with MUT hCOs showing markedly reduced expression levels (Figure 1P). Therefore, we subsetted progenitor and excitatory neuron clusters and evaluated the percentage of cells expressing cell type-specific markers (Figures S5H and S5J). Regardless of the genetic background and genotype, the majority of neural progenitors were double-positive for *FOXG1* and *LHX2*, confirming their dorsal telencephalic commitment, and equally expressed canonical markers known to represent radial glia (both aRG and oRG; *VIM* and *SLC1A3*), oRG (*HOPX* and *FAM107A*), and IPs (*EOMES* and *ELAVL4*)⁸⁵ (Figure S5I). A modest increase in oRG marker expression was noted in MUT hCOs, but given the known underrepresentation of oRGs in organoid systems,⁸⁶⁻⁸⁸ this difference should be interpreted cautiously. Excitatory neuron clusters also showed no consistent genotype-associated shifts in infragranular (*TBR1*, *TLE4*, *BCLL11B*, and *FEZF2*) or supragranular (*RORB*, *POU3F2*, *SATB2*, and *CUX2*) marker expression (Figure S5K). Immunostaining in patient 2042-derived hCOs confirmed equivalent numbers of BCL11B⁺ and SATB2⁺ neurons between genotypes (Figure S5L). Co-expression of infra- and supragranular-layer markers—expected due to immature laminar identity prior to GW 23 and the limited layering fidelity of organoids^{57,86,89}—was also similar across genotypes (Figure S5M). Together, these results highlight that *MeCP2* deficiency does not disrupt the temporal progression of neocortical fate acquisition.

MeCP2 regulates cortical excitatory neuron maturation

Given that MeCP2 is expressed at lower levels in progenitor cells than in neurons, we focused our analysis on the developmental progression of cells committed to the excitatory lineage. From IPs to neurons with infra- or supragranular identity, we observed an increased density of intermediate states along pseudotime in MUT hCOs, consistent with altered or protracted differentiation dynamics relative to WT (Figure 2A). To understand the implications of this shift in differentiation, we analyzed neuronal populations with defined layer identity and identified DEGs between *MeCP2* genotypes across genetic backgrounds. GO term enrichment revealed the upregulation of genes involved in axonogenesis, neuronal projection development, axon guidance, and synapse assembly in MUT neurons from both patients (Figure 2B). We then computed a pseudo-score along the excitatory lineage for the top three GO terms enriched in MUT neurons and found consistently higher scores in MUT hCOs compared to WT (Figure S6A). In agreement with this, immature

neurons also showed upregulation of genes associated with the development of projections and axons (Figure S6B). To identify genes consistently altered by *MeCP2* deficiency regardless of genetic background, we intersected DEGs between patients 2036 and 2042 (Figure 2C; Figure S6C). Shared upregulated genes in MUT hCOs included markers of early neuronal maturation and axonogenesis (*FRY*,⁹⁰ *MIR137HG*,^{91,92} *TUBB3*,⁹³ and *STMN1*⁹⁴), as well as regulators of synaptogenesis and cell adhesion (*LRRTM4*,⁹⁵ *ANK2*,⁹⁶ and *PCDH9*⁹⁷). Conversely, genes downregulated across both backgrounds included key transcription factors involved in cortical neuron differentiation and plasticity (*NEUROD2*,⁹⁸ *CREB5*,⁹⁹ and *POU3F2*^{80,100}). However, GO term enrichment analysis of downregulated genes revealed no overlap between the two genetic backgrounds (Figure 2D; Figure S6D). In patient 2036, enriched terms were primarily related to metabolic processes, particularly cholesterol biosynthesis known to support neurite outgrowth.¹⁰¹ In contrast, patient 2042 showed enrichment for terms associated with synapse organization and activity. Collectively, these results seem to point out that *MeCP2* deficiency results in broad transcriptomic dysregulation during early stages of cortical excitatory neuron maturation, with shared and patient-specific signatures.

Although upregulated genes largely pointed to accelerated programs of neuronal projection and axon development in MUT neurons, downregulated genes suggested regulatory impairments that may compromise neuronal morphological maturation and connectivity. Therefore, we next assessed the structural and functional features of cortical neurons in week 17 hCOs. We confirmed that *hsyn::eGFP*-labeled neurons co-localized with the excitatory lineage marker *NEUROD2* (Figure 2E), and morphological analysis revealed that MUT neurons were significantly more immature than WT neurons, exhibiting a smaller soma and fewer primary neurite branches (Figures 2E and 2F; Figures S6E and S6F). Although these differences were particularly pronounced in patient 2036, reduced neuronal arborization emerged as a consistent phenotype across both genetic backgrounds, aligning with pervasive neuroanatomical abnormalities observed in the brains of patients with RTT.¹⁰²⁻¹⁰⁴ To investigate neuronal function, we recorded spontaneous activity in week 17 hCOs from patient 2042 using GCaMP-based calcium imaging (Figure 2G; Videos S1 and S2). A striking observation was a significant increase in the inter-event interval (IEI) in MUT neurons (Figure 2H), indicating a marked reduction in overall spontaneous activity. Despite this, the burst duration, amplitude, and area under the curve (AUC) of calcium transients were significantly increased in MUT neurons (Figures 2I and 2J), potentially reflecting enhanced depolarization responses despite their lower firing frequency. Further supporting the notion of altered intrinsic neuronal excitability, single-cell profiling analysis also revealed downregulation of *GRIA4*, *KCNQ5*, and *NNAT* in MUT neurons (Figure 2K). *GRIA4*, a subunit of AMPA-type glutamate receptors, is essential for fast excitatory transmission¹⁰⁵; *KCNQ5* encodes a potassium channel involved in maintaining resting membrane potential and regulating excitability¹⁰⁶; and *NNAT* is associated with calcium signaling and neuronal differentiation.^{107,108} Together, these data suggest that *MeCP2* deficiency leads to dysregulated transcriptional programs that fail to support proper morphological and functional maturation of cortical excitatory neurons.

MeCP2 regulates cIN development

In the mouse cerebral cortex, MeCP2 expression is ~50% higher in GABAergic neurons than in non-GABA-releasing cells, and its global loss in inhibitory neurons recreates a near-complete RTT phenotype.^{30,109} GABA-releasing cINs are essential for cortical circuit formation yet remain underrepresented (or absent) in standard hCOs, reflecting the limited contribution of pallial progenitors to the total cIN pool.^{110,111} To overcome this limitation, we generated subcortical-like organoids and fused them with dorsal telencephalic counterparts, creating human telencephalic assembloids (hTAs).⁴³⁻⁴⁵ Following established protocols,^{43-45,112} concurrent sonic hedgehog (SHH) activation and WNT inhibition were used to induce medial ganglionic eminence (MGE) identity—a significant source of subcortical GABA-releasing cINs in the human telencephalon¹¹³ (Figure 3A; Figure S7A). To optimize MGE induction, we performed a dose-response curve of smoothened agonist (SAG) from day 10 to day 17 of differentiation (Figure S7B). A 100 nM concentration, as used in prior reports,^{43,44,114} did not efficiently ventralize progenitors. While we cannot exclude protocol-specific variables that may account for this discrepancy (e.g., timing of exposure and organoid density per well), recent studies have highlighted intrinsic variability across hPSC lines that can influence the efficiency of ventralizing cues.^{115,116} Indeed, expression of the MGE markers *NKX2-1* and *LHX6* was strongly dose dependent (Figures S7C-S7E), with a 500 nM SAG concentration needed to yield near-ubiquitous MGE identity and minimal presence of dorsal telencephalic progenitors across cell lines (Figure S8A). We refer to these ventralized organoids as human MGE organoids (hMGEOs). At week 5, hMGEOs robustly expressed *NKX2-1*, lacked the lateral ganglionic eminence (LGE) marker *GSX2*, and contained abundant *DLX2*⁺ IPs and/or immature cINs. By week 8, many of these cINs expressed mature GABAergic markers, including GABA and GAD67, confirming inhibitory lineage commitment (Figures S8B-S8D). To trace cIN migration, we developed an endogenous reporter using the mouse *Dlxi1/2b* enhancer^{44,117,118} via lentivirus-based genome insertion (Figures S8E-S8I). Upon fusion with hCOs, hMGEO-derived cINs exhibited extensive tangential migration into cortical tissue (Figure S8J). We further confirmed that *MeCP2* is expressed during the development of the inhibitory lineage, with single-cell RNA-sequencing data from the human subpallium during the first trimester of gestation (GW 9–12)¹¹⁹ revealing broad expression across all cell populations, and MeCP2 immunoreactivity in week 8 hMGEOs mirroring the pattern previously observed in hCOs (Figures S8K and S8L).

As *MeCP2* is expressed in the prenatal subpallium, we sought to investigate its functional role in the development of subcortical cINs. To this end, we generated hMGEOs from WT- and MeCP2-MUT hPSCs using the optimized ventralization protocol identified in our SAG dose-response analysis (Figures 3B and 3C). By week 3.5 of differentiation, WT and MUT hMGEOs were morphologically similar (Figure S9A) and showed robust MGE marker expression (Figures S9B and S9C), with broad *NKX2-1* expression and complete repression of the dorsal marker *PAX6* (Figure 3D; Figure S9C). After 5 weeks of differentiation, hMGEOs developed defined progenitor zones surrounded by *DLX2*⁺ IPs and/or immature cINs (Figure 3E), which expanded substantially by week 8 and were broadly marked by *Dlxi1/2b*-driven GFP (Figure 3F; Figure S9D). We next profiled the expression of genes associated with MGE specification (*NKX2-1* and *LHX6*), cIN development (*SOX6*, *DLX2*,

and *DLX5*), cIN migration (*ERBB4*, *MAFB*, and *CXCR4*), and GABA synthesis (*GAD1*) at week 8 of differentiation but found no significant differences between genotypes, though MUT hMGEs showed a slight elevation in *DLX5* and *ERBB4* expression (Figure S9E).

It has been previously suggested that there is altered expression of *DLX5* in RTT, under the assumption that MeCP2 forms a high-order chromatin loop at the *DLX5-DLX6* locus to control *DLX5* monoallelic expression.⁸ Given that MeCP2 is expressed in cINs in RTT hMGEs (Figure S9F) and that we observed a modest (~1.5-fold) increase in *DLX5* expression, we quantified DLX2⁺ and DLX5⁺ cINs in week 8 hMGEs. MUT organoids contained a higher proportion of both cell populations, independent of genetic background (Figures 3G and 3H; Figures S10A and S10B). To determine whether this increase in cINs could result from a loss of allelic expression at the *DLX5* locus, we screened our RTT cell lines for a common polymorphism located in the 3'-UTR of this gene (c.*163dupC^{8,120}) and found that three out of the four hPSC pairs were heterozygous. Both WT- and MUT-MeCP2 hMGEs exhibited biallelic *DLX5* expression, indicating that this gene is not imprinted and MeCP2 does not regulate its allelic expression state (Figure S10C). To further rule out relaxation of genomic imprinting in RTT during cIN development, we screened for polymorphisms in bona fide imprinted genes to serve as positive controls, namely *SNRPN*,¹²¹ *NDN*,¹²² and *PEG10*.¹²³ These genes were highly expressed in the fetal subpallium and maintained monoallelic expression regardless of genotype (Figure S10D). Taken together, these findings suggest that *MeCP2* deficiency leads to early cIN overproduction, independent of *DLX5* allelic regulation.

MeCP2 deficiency results in hypermotile cIN migration and cortical network hyperexcitability and hypersynchrony

In the mouse hippocampus, normal MeCP2 function is needed to maintain the asynchronous state of CA1 neurons, with inhibitory neurons required to enable neuronal circuits to resist hypersynchrony.³¹ Moreover, when *Mecp2* is selectively deleted from all inhibitory neuronal populations, mice show recurrent hyperexcitability discharges, albeit without any signs of electrographic seizures.³⁰ To examine MeCP2's role in developing cortical circuits upon the introduction of cINs, we fused pre-labeled Dlx1/2b::eGFP⁺ WT and MUT hMGEs with matched hCOs and tracked cIN integration into the cortical tissue (Figures 3I and 3J). Quantification of Dlx1/2b::eGFP⁺ coverage showed no genotype differences, indicating that MeCP2 loss does not impair cIN invasion (Figure 3K; Figures S11A and S11B).

Because *MeCP2* deficiency may result in subtle alterations in cIN developmental programs—rather than overt anatomical defects in their dispersion along the CP—we performed single-cell RNA sequencing on week 17 WT and MUT hTAs to disentangle how cIN dysfunction might arise in the absence of *MeCP2* within emerging cortical circuits. Unbiased clustering of 31,991 cells from six individual assembloids, derived from patient 2042, revealed eight clusters of telencephalic populations, with a clear segregation of cells belonging to dorsal and ventral telencephalic lineages (Figure 3L; Figure S11C). Cluster identities were assigned using canonical markers characteristic of mid-gestation telencephalic excitatory, inhibitory, and oligodendrocyte lineages (Figure 3M; Figure S11D), with this annotation showing strong correlation with *in vivo* fetal signatures (Figure

S11E). MUT hTAs showed a trend toward increased ventral cell proportions (Figure 3N; Figures S11F and S11G), consistent with the earlier cIN overproduction noted at week 8 in hMGEOs, though cINs were underrepresented overall—presumably because they are primarily localized at the organoid's surface and are thus the most sensitive to the single-cell dissociation process. A reduction in infragranular excitatory neurons was detected in MUT hTAs, a difference not previously seen in intact hCOs. This may reflect a cell-culture artifact introduced during the fusion process, which overlaps with peak stages of infragranular neuron generation.

Despite these limitations, and given the global reduction of *MeCP2* expression across all cell types (Figure S11H), we performed GO term enrichment analysis on DEGs from MUT excitatory neurons with defined layer identity and found a transcriptomic signature consistent with that observed in hCOs, including upregulation of genes involved in neuronal projection development and axonogenesis (Figures S11I and S11J). In MUT MGE-derived cINs, GO analysis revealed enrichment for genes associated with regulation of membrane potential—particularly potassium ion transport—as well as axonogenesis and neuronal migration (Figure 3O). Notably, several genes critical for interneuron maturation and function were upregulated, including those involved in tangential migration and guidance (*ERBB4*¹²⁴, *NRG3*¹²⁵ and *NRP2*¹²⁶), potassium channels that regulate neuron excitability (*KCNC2*¹²⁷ and *KCNIP4*¹²⁸), and GABA_A receptor subunits (*GABRA2* and *GABRB1*¹²⁹; Figure 3P; Figure S11K).

While *MeCP2*-deficient cINs exhibit an apparently normal chemoattraction toward cortical targets, their integration into local circuits relies on tightly timed tangential migration, which may be disrupted by the aberrant upregulation of guidance and migration genes (Figure 3O). To assess the role of *MeCP2* in cIN migration, we performed live imaging of intact hTAs to examine in detail the migratory behavior of *Dlx1/2b::eGFP*⁺ cells (Figure 4A), including the frequency and complexity of their movement (Figure S12A). Both WT- and MUT-*MeCP2* cINs exhibited the characteristic saltatory, start-stop motion in which the soma is propelled forward through successive nucleokinesis events¹³⁰; however, MUT cINs displayed increased nucleokinesis length (average inter-event displacement), higher event frequency, and an overall faster migration speed over the course of the recordings compared to WT cells (Figure 4B; Figures S12B and S12C; Videos S3 and S4). We also observed that MUT cINs were more predisposed to follow linear trajectories than WT cINs, exhibiting increased path directness and reduced turning angles, thereby avoiding multidirectional migratory behaviors (Figure 4B; Figures S12B and S12D).

We hypothesized that the aberrant migration observed in *MeCP2*-deficient cINs, coupled with dysregulation of transcriptional programs involved in cIN excitability (Figure 3O), could have severe consequences at the neural network level. To test this, we performed GCaMP-based calcium imaging of hTAs derived from patient 2042, approximately 2 months post-fusion—when cINs are expected to have begun integrating and modulating excitatory circuits.⁴⁴ Similar to our observations in hCOs, MUT-*MeCP2* neurons in hTAs displayed calcium traces with prolonged burst durations, as well as increased amplitude and AUC (Figures S12E and S12F). However, in contrast to the highly synchronous activity seen in hCOs regardless of genotype (Figure 2G), WT-*MeCP2* hTAs exhibited an overall

loss of synchrony following cIN integration, with reduced coordinated firing across the circuit (Figure 4C; Video S5). In turn, MUT-MeCP2 hTAs maintained synchronous activity and exhibited a substantial increase in neuronal firing frequency (Figure 4C; Video S6), reflected by a pronounced decrease in IEI compared to both WT-hTAs and MUT-hCOs (Figure S12G). To objectively assess this phenomenon, we computed the mean pairwise Pearson's correlation coefficient across multiple fields-of-view in both hCOs and hTAs (Figure 4D). This revealed decreased and increased correlated activity in WT- and MUT-hTAs, respectively, compared to their hCO counterparts (Figure 4E). Additionally, correlated activity was markedly higher in MUT-hTAs than in WT-hTAs, a finding that aligned with a reversed trend in the coefficient of variation of IEI across fields-of-view (Figure 4F). These findings suggest that while WT-MeCP2 cINs appropriately dampen synchronous activity in maturing circuits, *MeCP2*-deficient cINs fail to exert the same effect, leading to persistent hypersynchrony and increased network excitability.

DISCUSSION

MeCP2 is a multifactorial epigenetic regulator, and loss-of-function mutations in this gene underlie the genetic basis of RTT. Although RTT is often regarded as a postnatal disorder,⁵³ the MeCP2 protein is already widespread in the embryonic cerebral cortex,²⁷ with increased expression levels in neurons and varying degrees of protein accumulation depending on their birthdate.¹⁶ Our data demonstrate that *MeCP2* is expressed during mid-gestation in the human cortex, implicating MeCP2 in developmental processes preceding symptom onset. hCOs carrying RTT-associated *MeCP2* variants maintained the canonical inside-out sequence of excitatory neuron specification but exhibited broad dysregulation of transcriptional programs associated with neuronal maturation. Morphological and functional analyses of *MeCP2*-deficient excitatory neurons further indicated a delayed or incomplete maturation. A protracted or stalled differentiation trajectory was supported by pseudotime analysis along the excitatory lineage, and, together with the transient enrichment of progenitor over neuronal identities observed in 1-month organoids, this suggests that the upregulation of genes related to morphological development in MUT neurons may reflect a temporal lag relative to isogenic controls—even though corticogenesis proceeds normally. While additional temporal resolution will be needed to determine whether this reflects a true delay or instead indicates transcriptional dysregulation and the presence of compensatory mechanisms, our findings highlight MeCP2's importance to coordinate the tempo of neuronal maturation. This aligns with the view that MeCP2 acts as a global chromatin organizer to dampen transcriptional noise and coordinate genome-wide expression programs.^{1,29} Such temporal asynchrony parallels recent ASD organoid studies showing altered developmental trajectories and decoupling between transcriptional and functional programs.¹³¹⁻¹³³ Finally, the presence of both convergent and divergent transcriptional dysregulation across different genetic backgrounds is consistent with the heterogeneous expressivity seen in ASD populations.^{132,134} In our study, however, this effect cannot be separated from the influence of *MeCP2* variant type, which is itself known to modulate disease severity.⁶²

MeCP2 loss-of-function profoundly impacts the GABAergic system,^{30,109} affecting key components of inhibitory neurotransmission and contributing to excitation-inhibition

imbalance.^{30,135-137} By 8 weeks of differentiation, *MeCP2*-deficient hMGEOs already exhibited marked alterations in the inhibitory lineage development, notably a significant overproduction of DLX2⁺ and DLX5⁺ cINs. This increase in DLX5⁺ cells is not due to loss-of-imprinting,⁸ aligning with recent evidence that *DLX5/6* are biallelically expressed in the brain¹²⁰ and suggesting that other mechanisms are involved that merit further investigation. While cortical hyperexcitability in neurodevelopmental disorders is typically linked to reduced inhibition, cIN overproduction has paradoxically emerged as a recurrent feature in organoid models of idiopathic ASD and ASD-risk gene mutations,^{132,138,139} echoing our results. Although this surplus may be transient, its occurrence during critical periods of synaptic plasticity could induce long-lasting network miswiring with lasting consequences. Indeed, mouse models show that excess Parvalbumin⁺ interneurons from abnormal MGE SVZ proliferation persist into critical developmental windows, leading to long-term behavioral changes despite eventual normalization of cell numbers.¹⁴⁰

The importance of *MeCP2* in cIN development is further underscored by our findings that its loss leads to markedly abnormal tangential migration dynamics. In hTAs, *MeCP2*-deficient cINs migrating through the cortical compartment exhibited increased nucleokinesis displacement and frequency, along with higher overall speed—suggesting that *MeCP2* normally constrains cIN motility. Although cIN migration is governed by various cues—including chemorepellents from the MGE VZ,^{141,142} as well as chemoattractants that guide cells along defined migratory streams¹⁴³ and time their entry into the CP¹⁴⁴—the depolarizing action of GABA during early development is a potent motogenic signal.¹⁴⁵ In immature neurons, low expression of the chloride exporter KCC2 (*SLC12A5*), relative to the importer NKCC1 (*SLC12A2*), results in high intracellular chloride; as a result, GABA_A receptor activation leads to chloride efflux and membrane depolarization. As development progresses and KCC2 expression increases, the chloride gradient reverses, making GABA hyperpolarizing.¹⁴⁶ Notably, KCC2 is expressed earlier in cINs than in pyramidal neurons and limits their motility.¹⁴⁵ Although KCC2 is reduced in *Mecp2*-null mouse brains,¹³⁷ our hTA neurons predominantly express NKCC1 regardless of genotype (Figure S12H), suggesting that GABA remains depolarizing at this stage. However, elevated expression of potassium channels and ligand-gated GABA_A receptor subunits in MUT cINs may amplify GABAergic depolarization, driving hypermotility. Still, non-cell-autonomous effects from the dorsal compartment, which provides extrinsic guidance cues,^{144,145,147} may also contribute to the expressivity of these phenotypes and warrant further investigation. In addition to altered speed and displacement, MUT cINs also showed fewer directional changes, avoiding the exploratory, multidirectional paths typically seen in the marginal zone of the rodent cortex.¹⁴⁸ This behavior facilitates even cIN dispersion across the CP, disruption of which has been linked to impaired inhibitory tone and increased seizure susceptibility.^{144,149,150}

Our findings of reduced spontaneous activity in *MeCP2*-deficient cortical neurons within hCOs are consistent with previous studies in postnatal *Mecp2*-null mice that reported reduced firing frequency before the emergence of overt behavioral symptoms.¹³⁵ Despite overall hypoactivity, we also detected increased burst dynamics in MUT excitatory neurons, suggesting heightened excitability amid impaired baseline activity. These observations align with morphological immaturity and a disruption of activity-dependent processes. Indeed,

CaMKII-dependent phosphorylation of MeCP2, triggered by calcium influx, is known to regulate dendritic architecture and activity-dependent gene expression,¹⁵¹ implying that *MeCP2* loss may decouple structural and functional maturation. Complementarily, *MeCP2*-deficient hTAs displayed persistent hypersynchronous network activity, in contrast to WT assembloids, which showed activity decorrelation upon cIN integration. Similar hypersynchrony has been observed in RTT hTAs,⁵¹ but in our model, it arose spontaneously without pharmacological manipulation. Such a pattern resembles the epileptiform abnormalities common in patients with RTT¹⁵² and may reflect delayed maturation of inhibitory signaling. During early development, GABA acts depolarizingly to drive synchronous bursts, with later transitions toward sparse, desynchronized activity relying on the GABAergic switch to hyperpolarization.¹⁵³⁻¹⁵⁵ Persistent hypersynchrony in *MeCP2*-deficient neurons may thus signal a disruption in this developmental trajectory, potentially linked to a more pronounced depolarizing effect during early stages of neurodevelopment.

Overall, our work builds on earlier hPSC-based 2D models of RTT and provides a longitudinal, lineage-resolved view of how *MeCP2* deficiency impacts telencephalic development. In doing so, it reinforces the notion that MeCP2 plays critical roles in the prenatal brain and underscores the importance of further investigating its role during early developmental windows of heightened plasticity. Importantly, given that previous animal studies have shown that restoration of MeCP2 in post-symptomatic models can reverse neurological defects,¹⁵⁶ our findings support the possibility that early developmental insights may directly inform therapeutic strategies.

Limitations of the study

The use of cerebral organoids to study genetic variants associated with neurodevelopmental disorders carries inherent limitations. While these models recapitulate key features of early human neocortical development up to mid-gestation, many disorders, including RTT, manifest postnatally, and how accurately organoids reflect later stages of development remains debatable.^{47-49,157} Although our analysis revealed no major shifts in excitatory lineage cell type proportions—consistent with previous studies that conducted a more superficial profiling⁵¹—other reports have identified changes in specific populations, such as IPs and pre-plate neurons.⁵⁰ These discrepancies may reflect differences in model construction, including protocol choice and patterning fidelity.⁴⁸ Notably, congenital brain malformations are not characteristic of classical RTT but occur in the Rolando variant caused by *FOXG1* mutations.¹⁵⁸ Future studies should also resolve whether potential changes in cell-type composition due to *MeCP2* dysfunction are transient, though current organoid models do not capture later gestational stages.

Furthermore, the extent to which transcriptomic alterations observed in organoids parallel *in vivo* biology remains uncertain. Postmortem single-cell profiling studies in patients with RTT show limited overlap with our enrichment results,¹⁵⁹ suggesting that our findings reflect early prenatal dysregulation. While some developmental processes, such as tangential cIN migration, extend into early infancy—particularly in the frontal lobe¹⁶⁰—the overall translatability of these phenotypes to postnatal stages remains unclear. Another caveat concerns the timing of MeCP2's functional relevance. RTT symptoms coincide

with the accumulation of non-CG methylation (mCH),^{1,2,161} yet the mid-fetal cortex and, by extension, cerebral organoids largely lack this epigenetic mark.^{162,163} Thus, the transcriptional changes reported here are likely mCH independent, raising questions about how MeCP2 modulates gene expression prenatally. The broad nature of our functional assessments should also be interpreted cautiously, as the patient line used carried a variant associated with a higher seizure risk—capturing the severe end of the phenotypic spectrum, similar to *MeCP2*-null mouse models—even though, in RTT, a greater diversity of neuronal dysfunction predisposition exists.¹⁶⁴

Finally, as RTT is an X-linked disorder, the mosaicism resulting from XCI in female brains is not captured in our fully WT or MUT organoids. Future work employing mosaic or chimeric organoid systems¹⁶⁵ could better model cell-autonomous effects, though careful attention to the WT:MUT cell ratio—modulated *in vivo* by XCI skewing⁶⁵—will be critical, as this could significantly influence the severity and nature of observed phenotypes.

STAR★METHODS

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Experimental cell lines: Control hPSC lines: The hESC line H9 (WA09, WiCell) and the hiPSC lines KOLF2.1J,¹⁶⁶ WT83.06,¹⁶⁷ and WT.Fc02 were used as control cell lines in this study. The KOLF2.1J line was subcloned from WTSli018-B-12, which was reprogrammed using Sendai vectors, the WT83.06 line was reprogrammed using lentivirus vectors, and WT.Fc02 line was reprogrammed using retrovirus vectors. All cell lines used in this study were routinely checked by karyotype and CNV arrays to avoid genomic alterations in culture.

Experimental cell lines: RTT hPSC lines: RTT hiPSCs were obtained from the Rett Syndrome Research Trust (RSRT) iPSC Collection deposited at the Coriell Institute for Medical Research and were derived from blood lymphocytes and reprogrammed at the Harvard Stem Cell Institute with Sendai virus vectors (CytoTune-iPS 2.0 Sendai Reprogramming Kit, Invitrogen, A16518). The RTT hiPSC lines used in this study carry mutations in the *MeCP2* locus: subject 2036 harbors a C>T missense mutation at nucleotide 397 (c.397C>T), resulting in a substitution of a cysteine in place of an arginine at codon 133 (p.R133C); subject 2042 harbors a C>T nonsense mutation at nucleotide 763 (c.763C>T), resulting in a premature stop codon in place of an arginine at codon 255 (p.R255X); subject 2047 harbors a C>T missense mutation at nucleotide 916 (c.916C>T), resulting in a substitution of a cysteine in place of an arginine at codon 306 (p.R306C). Patient 2036 cells were isolated from a 4-year-old female with a Clinical Global Impression of Severity score of 4, no clinical history of seizures, and first concerns noted by parents at 13 months of age due to a missed milestone in speech development. Patient 2042 cells were isolated from a 7-year-old female with a Clinical Global Impression of Severity score of 6, a clinical history of near-daily seizures, and first concerns noted by parents at 10 months of age due to a missed milestone of pulling to stand. Patient 2047 cells were isolated from a 13-year-old female with a Clinical Global Impression of Severity score of 5, a clinical history of monthly seizures, and first concerns noted by parents at 9 months of age due to absence

of weight bearing. As X chromosome inactivation occurs randomly during development, females with RTT present somatic mosaicism regarding the *MeCP2* mutation, with cells expressing either the mutant (MUT) or wild-type (WT) allele. After reprogramming, the inactive X chromosome (Xi) remains inactive,⁶⁴ which allows the isolation of MUT-*MeCP2* and WT clones from the same subject, the latter serving as an isogenic control.

A fourth *MeCP2*-deficient line (denominated here as 2982) was used in this study that was derived from the H9 hESC cell line via CRISPR/Cas9 introduction of an homozygous deletion at nucleotide 246 (c.246delA), resulting in a sequence frameshift from codon 82 and a premature stop codon (p.K82Nfs*43), as previously described.⁶⁸ hPSC line experimental allocation is detailed in Table S1.

Confirmation of *MeCP2* allelic status: Confirmation of *MeCP2* monoallelic expression in RTT cell lines was routinely performed with real-time PCR (RT-PCR) at the hiPSC stage and/or throughout differentiation, as described below. Throughout the duration of this study, biallelic expression of *MeCP2* (suggestive of Xi erosion⁶⁷) was never detected in any hPSC clone.

METHOD DETAILS

hPSC culture: hPSC lines were propagated and maintained in feeder-free conditions in six-well (Corning, 3516) Matrigel-coated plates (Matrigel hESC-Qualified Matrix, Corning, 354277) and cultured with mTeSR Plus medium (StemCell Technologies, 100–0276), with daily medium changes or every other day, as instructed by the manufacturer. Cultures were monitored daily under phase contrast to assess cell quality. Healthy and good-quality hPSC colonies were classified as having a well-defined border, tight cellular packing, and a high nucleus-to-cytoplasm ratio. Subculturing of hPSCs was performed when cell confluency reached 70–80% with ReLeSR Reagent (StemCell Technologies, 100–0484). hPSC subculturing was typically undertaken every 3–5 days. Cells were passaged two to three times before differentiation. All cell lines used in this study were routinely tested for mycoplasma. hPSC lines were kept in a humidified 5% CO₂ incubator at 37°C.

Plasmid construction and generation of *Dlxi1/2b:eGFP* reporter hPSC lines:

Dlxi1/2b:eGFP reporter lines were generated by infecting hPSC with LV-*Dlxi1/2b:eGFP*-WPRES-Puro followed by antibiotic selection. The transfer vector used for lentiviral production is a variation of the pLenti-*Dlxi1/2b:eGFP* vector that contains a puromycin resistance gene (*pac* gene, which encodes puromycin N-acetyl-transferase). The pLenti-*Dlxi1/2b:eGFP* vector (gift from J. L. Rubenstein and S. P. Paşca) contains part of the mouse *Dlx1* and *Dlx2* intragenic enhancer (enhancer 422, coordinates - chr2: 71,374,047–71,374,552 [mm9]), a β -globin minimal promoter, and the eGFP coding sequence in tandem, as previously described.^{44,117,118} The puromycin resistance gene is under the control of the PGK promoter and was retrieved from pLenti-CMV::GFP-WPRES-Puro (gift from E. Campeau and P. Kaufman through addgene, plasmid 17448). Both the *Dlxi1/2b:eGFP* vector and the puromycin resistance gene were PCR amplified using PrimeSTAR GXL DNA high-fidelity polymerase (Takara, R050A), followed by treatment of the PCR products with DpnI (NEB, R0176S) and purification using the Monarch PCR

& DNA cleanup kit (NEB, T1030S). The puromycin resistance gene was cloned into the pLenti-Dlx1/2b:eGFP vector after the WPRE sequence with the NEBuilder HiFi DNA Assembly Cloning Kit (NEB, E5520S) using 0.1 pmol of vector and a 1:2 DNA molar ratio of vector:insert. Confirmation of insert integration was carried out via standard colony PCR. The integrity of the resultant vector construct was verified by whole plasmid sequencing.

hPSCs were cultured in a 6-well plate and were grown until 20–30% confluency (~0.2–0.5x10⁶ cells per well) before infection with LV-Dlx1/2b:eGFP-WPRE-Puro (obtained from VectorBuilder). Infection was carried out in 1 mL of mTeSR Plus containing 5 µg/mL polybrene and with a MOI of 1. Approximately 6h post-infection, another 1 mL of medium was added per well. Cells were incubated with virus for 48h total, and the medium was replenished on day 3 post-infection. When cells reached 70–80% confluency (around day 4 post-infection), 1.2 µg/mL of puromycin (InvivoGen, ant-pr-1) in mTeSR Plus was supplemented for 4–5 days, with daily medium changes. Surviving colonies were manually picked, expanded, and cryopreserved for use. Transgene genomic integration was verified by PCR targeting *eGFP* and *pac*. Puromycin resistance of cryopreserved cells was rechecked prior to differentiation.

Generation of hCO and hMGEO from hPSC: hCO differentiation from hPSCs was based on previous reports,^{59,168,169} with modifications. On day 0 of differentiation, if any spontaneous differentiation was detectable in hPSC cultures, such cells were removed by selective scraping. To generate a single-cell suspension, hPSCs were incubated with Accutase (Life Technologies, A11105-01) at 37°C for 7 min. To obtain uniformly sized hPSC aggregates, about 3 x 10⁶ cells were seeded per AggreWell800 plate well (StemCell Technologies, 34815) in mTeSR Plus medium supplemented with 10 µM ROCK inhibitor Y-27632 (Tocris, 1254). The plate was then centrifuged at 200 x g for 3 min to capture the cells in the microwells and incubated overnight at 37°C with 5% CO₂. On day 1 of differentiation, hPSC aggregates of about 300–400 µm were dislodged from the microwells, resuspended in Neural induction medium 1 (NIM1), transferred to an ultra-low attachment 6-well plate (Corning, 3471), and placed on an orbital shaker at 90 rpm. NIM1 contained DMEM-F12 with HEPES (Life Technologies, 11330-032), 15% v/v Knock-Out Serum Replacement (Life Technologies, 10828-028), 1% v/v MEM-NEAA (Life Technologies, 11140-050), 1% v/v GlutaMAX (Life Technologies, 35050-061), 100 µM β-Mercaptoethanol (Life Technologies, 31350-010), 100 nM LDN-193189 (Sigma-Aldrich, SML0559), 10 µM SB-431542 (Stemgent, 04-0010), and 2.5 µM XAV-939 (Sigma-Aldrich, X3004). Medium was replenished daily until day 6. On day 6, organoids were transferred to NIM2, which has a similar composition to NIM1 except that Knock-Out Serum Replacement was swapped with 1% v/v N2 supplement (Life Technologies, 17502-001). NIM2 was supplemented with inhibitors on day 6. Medium containing low bFGF (4 ng/mL; Life Technologies, PHG0261) was changed on days 8 and 9, but patterning factors were left out. On day 10, batches in which over 90% of organoids formed successfully were kept for further experiments. Quality criteria included neural aggregate size above 400 µm, round morphology with a regular border, and the appearance of peripheral tissue clearing suggestive of successful neuroepithelium induction. From day 10 to day 24, organoids were cultured in Neural differentiation medium 1 (NDM1). NDM1 corresponds to a 1:1 mixture

of DMEM-F12 with HEPES and Neurobasal medium (Life Technologies, 21103-049), supplemented with 0.5% v/v N2 supplement, 2% v/v B-27 supplement without vitamin A (Life Technologies, 12587-001), 0.5% v/v MEM-NEAA, 1% v/v GlutaMAX, and 0.025% v/v insulin (Sigma-Aldrich, I9278). NDM1 was further supplemented with 20 ng/mL bFGF for the first 7 days with daily medium changes, and a combination of 20 ng/mL bFGF and 20 ng/mL EGF (PeproTech, 100-15) was added to the medium for the remaining 8 days with medium changes every other day. From day 25 onwards, organoids were cultured in NDM2, which has a similar composition to NDM1 except that B-27 supplement containing vitamin A was used instead (Life Technologies, 17504-001), and the medium was replenished thrice a week. Starting around weeks 4–5 of differentiation, NDM2 started to be supplemented with 1% v/v Matrigel (Corning, 354234) to aid cortical plate formation and maintain organoid integrity for long-term culture. From day 80 onwards, medium was changed twice a week.

The differentiation of hMGEO from hPSCs was similar to hCO differentiation, with the following modifications during neural patterning. On day 6, organoids were cultured in NIM2 containing low bFGF and 1.25 μ M XAV-939 (without dual-SMAD inhibitors). Medium was changed on day 8, and the aforementioned factors were replenished. From day 10–24, NDM1 was supplemented with 20 ng/mL bFGF/FGF2, with the same feeding regimen as for hCO. Additionally, 500 nM SAG (Selleckchem, S7779) and 1.25 μ M XAV-939 were supplemented to NDM1 from day 10 to day 17, with daily medium changes.

Medium seeding volume in Aggrewell800 plate format was 1.5 mL per well, and medium feeding volume in 6-well plate format was 3 mL per well. At later stages of differentiation, feeding volume in 6-well plate format for hCOs was 4 mL per well. On day 1, a 1:1 split ratio was used to transfer organoids from Aggrewell800 to 6-well plate format. Organoids were subsequently split 1:2 on day 17. Past day 25, no more than 20 organoids were kept per well of a 6-well plate. For hCOs, from day 50–60 onwards no more than 8–12 organoids were kept per well, and past day 80 no more than 4–5 organoids were kept per well. All the differentiation media described above were prepared fresh on a weekly basis and kept at 4°C. Penicillin-streptomycin (Life Technologies, 15140122) was added at 1% v/v to the medium during neural induction, and antibiotic-antimycotic was added at 1% v/v to the differentiation medium (Life Technologies, 15240062). Patterning factors and Matrigel were added to the medium prior to use.

Generation of telencephalic assembloids: hTA were generated via controllable fusion of intact hCO and hMGEO, as previously described,⁴⁴ with modifications. In brief, a single hCO and a single hMGEO were positioned in proximity at the bottom of a 1.5 mL Eppendorf tube containing 1 mL of NDM2. Around two-thirds of the medium was replaced on days 3 and 5 without disturbing the organoids, and the fusions were transferred to a 12-well plate on day 7 and cultured under orbital shaking at 90 rpm. Fused organoids were initially cultured individually, and once the interface was stable (around 2 weeks after fusion), up to 5 assembloids were cultured per well of a 6-well plate under agitation at 90 rpm. hTAs were long-term cultured in NDM2 supplemented with 1% v/v Matrigel, and medium was changed two to three times a week depending on the stage of the differentiation.

Genomic DNA isolation and genotyping: Cells were lysed overnight at 56 °C (lysis buffer: 100 mM NaCl, 10 mM Tris pH 8.0, 25 mM EDTA pH 8.0, 0.5% v/v SDS, with 0.2 µg/µL Proteinase K (Invitrogen, EO491)) and genomic DNA was isolated via conventional phenol:chloroform:isoamyl alcohol (Sigma-Aldrich, P2069-100ML) extraction. Genomic DNA was stored at – 20°C until further use. For genotyping, PCR was performed with 15 ng of genomic DNA using Xpert Fast Hotstart DNA polymerase (Xpert Fast Hotstart Mastermix (2X), grisp, GE35.5001), with the following cycling parameters: 95°C for 3 min, followed by 35 cycles of 30 s at 95°C, 30 s at T_m, and 30 s at 72°C, and then a final step at 72°C for 5 min. Primers and the respective T_m used are listed in Table S2. After band excision and/or PCR product clean-up, amplicons were sent for Sanger sequencing. Chromatograms were visualized and analyzed with SnapGene Viewer v.5.3.2 software.

RNA extraction: Total RNA was extracted from cells and organoids using TRIzol Reagent (Invitrogen, 15596026-100ML), and up to 5 µg of RNA was subsequently treated with DNaseI (Roche, 04716728001), according to the manufacturer's instructions. RNA was stored at – 80°C until further use.

RT-PCR and allelic expression analysis: 500 ng or 1 µg of DNaseI-treated RNA was reverse-transcribed into cDNA with QuantiTect Reverse Transcription Kit (QIAGEN, 205313). cDNA was stored at – 20°C until further use. To analyze allelic expression in WT and MUT cells, approximately up to 50 ng of cDNA was amplified by RT-PCR using Xpert Fast Hotstart DNA polymerase (Xpert Fast Hotstart Mastermix (2X), grisp, GE35.5001), with the following cycling parameters: 95°C for 3 min, followed by 35 cycles of 30 s at 95°C, 30 s at T_m, and 30 s at 72°C, and then a final step at 72°C for 5 min. Primers and the respective T_m used are listed in Table S3. After PCR product clean-up, amplicons were sent for Sanger sequencing. Chromatograms were visualized and analyzed with SnapGene Viewer v.5.3.2 software.

RT-qPCR: cDNA was generated as described for RT-PCR analysis. Quantitative RT-PCR (RT-qPCR) was performed using RT² SYBR Green Fluor qPCR Mastermix (QIAGEN, 330523) on a CFX384 Touch Real-Time PCR detection system with a CFX Maestro software (Bio-Rad; version 2.3), with the following cycling parameters: 95°C for 10 min, followed by 40 cycles of 15 s at 95°C and 1 min at 60°C. Amplification and denaturation curves were analyzed to determine the amplification of a single amplicon. Primers used are listed in Table S4. All RT-qPCR reactions were run in triplicate. The relative expression of each gene was computed via the $2^{-\Delta CT}$ method using *TBP* or *RPL27* as an endogenous control.

Cryosectioning: Organoids were fixed in 4% formaldehyde (ThermoFisher Scientific, 28908) overnight at 4°C, with gentle agitation. After three 10 min washes in PBS, organoids were immersed in 30% sucrose (Sigma-Aldrich, S0389) and allowed to sink for 24-48h at 4°C. Then, organoids were incubated in a 10% sucrose/7.5% gelatin (Sigma-Aldrich, G1890) solution overnight at 37°C, embedded in the same sucrose/gelatin solution the next day, and allowed to solidify at 4°C overnight. Samples were flash-frozen in a bath of 2-methylbutane (Sigma-Aldrich, MX0760) super-cooled to a temperature of about –60°C by

dry ice. Cryoblocks were stored at -80°C until further processing. Cryoblock sectioning was carried out in a cryostat (Leica, CM3050 S) at $20\text{ }\mu\text{m}$ thickness, sections were collected onto Fisherbrand Superfrost Plus microscope slides (FisherScientific, 12-550-15), and stored at -80°C until further use.

Immunofluorescence staining

Adherent cells: Cells grown on glass coverslips were fixed with 4% w/v paraformaldehyde (ThermoFisher Scientific, 28908) for 10 min at RT, followed by three 10 min washing steps in PBS. Cells were permeabilized with 0.1% v/v Triton X-100 in PBS for 15 min at RT, blocked with 5% v/v BSA for 1h at RT, and primary antibodies were diluted in blocking buffer and incubated overnight at 4°C or for 2h at RT. Then, secondary antibodies were diluted in blocking buffer and incubated for 30–45 min at RT. Nuclei were counterstained with 4',6-diamidino-2-phenylindole (DAPI, $1.0\text{ }\mu\text{g/mL}$; Sigma-Aldrich, D9564), and coverslips were mounted in ProLong Gold antifade reagent (Invitrogen, P36930). Slides were imaged on an Andor Dragonfly 200 spinning disc confocal microscope (Oxford Instruments). Antibodies and the respective dilutions used are listed in Table S5.

Cryosectioned organoids: Slides were degelatinized for 30–45 min in PBS at 37°C . Then, slides were rinsed with PBS before permeabilization with 0.2% Triton X-100 in PBS for 1h at RT. Sections were blocked with blocking solution consisting of 5% BSA in PBS with 0.1% Tween 20 (PBST) for 2h at RT. Primary antibodies were diluted in blocking solution and applied to the sections for 3.5h at RT or overnight at 4°C . After washing with PBST, secondary antibodies were diluted in blocking solution and incubated for 1h at RT. Finally, sections were washed with PBST, stained with DAPI, and mounted in ProLong Gold antifade reagent. Antibodies and the respective dilutions used are listed in Table S5. Slides were imaged on an Andor Dragonfly 200 spinning disc confocal microscope (Oxford Instruments).

Organoid and assembloid clearing: The Clear, Unobstructed Brain/Body Imaging Cocktails and Computational analysis (CUBIC) method was used to optically clear hCOs and hTAs, as previously described.^{46,170,171} hCOs and hTAs were fixed in 4% formaldehyde overnight at 4°C , with gentle agitation. Then, samples were washed with PBS (2h X3, RT) before incubation with 50% Tissue-Clearing Reagent CUBIC-L (TCI, T3740) at 37°C overnight. On the next day, samples were incubated with 100% CUBIC-L for 1d at 37°C . After washing with PBS (2h X3, 37°C), samples were incubated with RedDot2 Far-Red Nuclear Stain (biotium, 40061; 1:150 diluted in PBS with 500 mM NaCl) at 37°C overnight. Samples were then washed with PBS (2h X3, 37°C) followed by incubation with HEPES-TSB (10 mM HEPES, 10% Triton X-100, 200 mM NaCl, 0.5% BSA, 0.01% sodium azide) at 37°C for 2h. hCOs and hTAs were subsequently stained with anti-GFP antibody (GeneTex, GTX13970, 1:1,000) for 2d at 37°C . After primary antibody incubation, samples were washed with 10% Triton X-100 in PBS (2h X2, 37°C) and HEPES-TSB (2h X1, 37°C), and then incubated with secondary antibody (Alexa Fluor 488-conjugated AffiniPure Donkey anti-Chicken IgY, Jackson ImmunoResearch, 703-545-155; 1:300 dilution) for another 2d at 37°C . Antibodies were diluted in HEPES-TSB buffer. Following secondary antibody incubation, samples were washed with 10% Triton X-100 in PBS (30 min X2,

37°C) and PBS (1h X1, 37°C), followed by incubation with 100% CUBIC-R+(M) (TCI, T3741) at RT for 1d. Optically cleared samples were imaged on an Andor Dragonfly 200 spinning disc confocal microscope (Oxford Instruments).

RNA fluorescence *in situ* hybridization for *Xist*: The RNA-FISH protocol was adapted from the Stellaris (Biosearch Technologies, Hoddesdon United Kingdom) protocol for adherent cells. Cells were grown on glass coverslips and fixed with 4% formaldehyde at RT for 10 min, followed by two 10 min washing steps in PBS. Fixed cells were permeabilized with 0.25% v/v Triton X-100 for 10 min. Then, cells were washed with Stellaris RNA-FISH Wash Buffer A (Biosearch Technologies, SMF-WA1-60) containing 10% v/v formamide (Sigma-Aldrich, S4117) for 5 min at RT (wash buffer A), followed by incubation at 37°C for 4h (to overnight) with 125 nM human *Xist* probes labeled with Quasar570-dye (Biosearch Technologies, SMF-2038-1) prepared in Stellaris RNA-FISH hybridization buffer (Biosearch Technologies, SMF-HB1-10) containing 10% v/v formamide. After hybridization, cells were washed with wash buffer A for 30 min at 37°C, and the nuclei were stained with DAPI for 30 min at 37°C. Cells were then washed with Stellaris RNA-FISH Wash Buffer B (Biosearch Technologies, SMF-WB1-20) for 5 min at RT, and coverslips were mounted in ProLong Gold antifade reagent. Slides were imaged on an Andor Dragonfly 200 spinning disc confocal microscope with a 63X oil-immersion objective.

Interneuron migration imaging: For live imaging of interneuron migration in hTAs, day 50–55 hMGEOs were infected with LV-Dlxi1/2b:eGFP and then assembled with hCOs 7–8 days later, as described above. To virally label hMGEOs, day 50–55 hMGEOs were transferred to a 1.5 mL Eppendorf tube containing 200 µL NDM2 with 5 µg/mL polybrene and incubated with $\sim 2.5 \times 10^7$ TU of LV-Dlxi1/2b:eGFP (lentiviral preparation obtained from VectorBuilder) overnight at 37°C with 5% CO₂. The following day, 800 µL of fresh NDM2 was added, and hMGEOs were incubated with virus for an additional 24h. The next day, hMGEOs were transferred back to a 6-well plate and kept under agitation until assembly.

hTAs were transferred to a 96-well glass bottom microplate (ibidi, 89627; one assembloid per well) in 300 µL NDM2 and incubated in an environmentally controlled chamber at 37°C and with 5% CO₂ for 1h before imaging on an Andor Dragonfly 200 spinning disc confocal microscope (Oxford Instruments), and up to 7 assembloids were imaged in a given session. hTAs were imaged with a 10X air objective (2048× 2048-pixel resolution, 2308.60 × 2309.80 µm field-of-view) at a depth of 500-700 µm and at a rate of 20 min per frame for a total of 18-20h.

GFP labeling of cortical neurons: hCOs were infected with AAV7m8-hsyn1-eGFP (viral preparation obtained from Unitat de Producció de Vectors, Universitat Autònoma de Barcelona) around day 110 of differentiation to label excitatory cortical neurons. Up to 4 organoids were infected per well of a 6-well plate in 3 mL of NDM2 containing 1×10^{10} vg, under agitation. Organoids were incubated with virus for approximately 6h, after which another 1 mL of fresh NDM2 was added per well. Medium was fully replaced 72h after

infection. After 7–10 days post-infection, organoids were fixed and optically cleared for analysis.

Calcium imaging: For live calcium imaging, the genetically encoded calcium reporter GCaMP6s was introduced into hCOs and hTAs around day 100 of differentiation via infection with AAV7m8-hSyn1-GCaMP6s-P2A-nls-dTomato (gift from Jonathan Ting through addgene, plasmid 51084; viral preparation obtained from VectorBuilder). AAV infection was carried out as aforementioned and up to 5 organoids were infected per well with 5×10^{10} vg (except the medium formulation used was the one mentioned below). Prior to recordings, hCOs and hTAs were gradually transitioned to BrainPhys neuronal medium (StemCell Technologies, 05790) starting around day 80 (week 11, 25%; week 12, 50%; week 13, 75%; week 14, 100%). BrainPhys neuronal medium was supplemented with 1% v/v CD lipid concentrate (Life Technologies, 11905031), 1:147 20% glucose solution (Life Technologies, A2494001), 1% v/v N2 supplement, 2% v/v B-27 supplement with vitamin A, 20 ng/mL BDNF (PeproTech, 450-02), 20 ng/mL GDNF (PeproTech, 450-10), 1 mM cAMP (Sigma-Aldrich, D0627), 200 μ M L-ascorbic acid (Sigma-Aldrich, A4403), and 1% v/v antibiotic-antimycotic. Medium was changed twice a week. On the day of the recordings (around 20 days after infection, week 17 of differentiation), organoids were transferred to a 96-well glass bottom microplate (one hCO or hTA per well) in 300 μ L of BrainPhys Imaging Optimized Medium (StemCell Technologies, 05796) with supplements (BDNF, GDNF, cAMP, and ascorbic acid were left out) and incubated in an environmentally controlled chamber at 37°C and with 5% CO₂ for 30 min before imaging on an Andor Dragonfly 200 spinning disc confocal microscope (Oxford Instruments). Calcium transients were recorded using a 40X glycerol-immersion objective (2048 × 2048-pixel resolution, 154.40 × 154.40 μ m field-of-view) and at a frame rate of 3 Hz.

Imaging data analysis

Organoid size: Brightfield images of week 3.5 organoids were acquired in an EVOS M5000 imaging system (Invitrogen) with a 4X air objective. Measurements of the organoid area and diameter were performed in Fiji software (ImageJ, v1.54k) by manually outlining the organoid's border with the “Freehand” tool.

Ventricular zone size and thickness: Week 5–6 old hCOs were sectioned and immunostained for SOX2 and FOXG1 alongside DAPI. Only cryosections near the middle of the organoid were used for quantification. The entirety of the organoid section was imaged with a 20X air objective (1024 × 1024-pixel resolution). The Ventricular zone (VZ) was defined by exclusive SOX2 immunoreactivity and neural tube-like morphology. Only VZs that also stained positive for FOXG1 were analyzed. z stack images were acquired and projected with maximum intensity in Fiji. The “Line” tool was used to measure the distance between the apical membrane and the outer loop, and an average of three measurements was considered to define the VZ thickness. The “Freehand” tool was used to trace the outer and inner loops, and the area of the VZ was defined as the subtraction of these two areas.

Density of BCL11B⁺ and SATB2⁺ cells: Week 17 old hCOs were sectioned and immunostained for BCL11B and SATB2. Cryosections in the vicinity of the organoid's

midline (± 300 – $400\ \mu\text{m}$) were used for quantification (at this stage of differentiation, organoids have an average size of 3–4 mm and, therefore, have an unhealthy core, which makes it more difficult to obtain clean sections as the center is approached during cryosectioning). z stack images were acquired with a 40X glycerol-immersion objective (2048×2048 -pixel resolution) and projected with maximum intensity in Fiji for analysis. From the outermost organoid surface (pia), regions of $500\ \mu\text{m}$ width spanning the full thickness of the cortical plate (CP) were randomly selected and cropped with the “Rectangle” tool in the direction of maximal radial distance. For the quantification of marker-positive nuclei, deep learning-based segmentation was performed via the “StarDist” plugin in Fiji using the “Versatile (fluorescent nuclei)” built-in model (<https://github.com/stardist/stardist>). The number of marker-positive nuclei was normalized to the area of the CP selection.

Cortical neuron distribution: Week 17 old hCOs sections were immunostained for BCL11B and SATB2, and random cortical structures were imaged with a 40X glycerol-immersion objective (2048×2048 -pixel resolution) and projected with maximum intensity in Fiji. From the pial surface, columns of $100\ \mu\text{m}$ width spanning the full thickness of CP layer were cropped with the “Rectangle” tool in the direction of maximal radial distance and used for analysis. The positions of all BCL11B⁺ and SATB2⁺ nuclei were separately marked using the “Cell Counter” plugin in Fiji. To measure nuclei’s laminar positioning, their y-coordinates were normalized to the full thickness of the CP selection, and the frequency distribution of their relative vertical positions was computed across 11 evenly distributed bins.

Cortical neuron morphology: Week 17 old hCOs labeled with AAV7m8-hsyn1-eGFP were optically cleared using the CUBIC method (without nuclear staining), as aforementioned, and random regions of the organoid were imaged (~ 200 – $300\ \mu\text{m}$ volume) with a 40X glycerol-immersion objective (2048×2048 -pixel resolution). To estimate the size and shape of the soma, frames from the bottom to the top of the cell body were selected, projected with maximum intensity in Fiji, and the “Freehand” tool was used to trace the soma to compute its area and circularity. Neuronal dendritic complexity was estimated by counting the number of primary branches for each neuron, defined as dendrites whose main root comes from the neuron’s soma.

Quantification of DLX2⁺ and DLX5⁺ cortical interneurons: Week 8 old hMGEOs were sectioned and immunostained for DLX2 and DLX5 alongside DAPI. Only cryosections near the middle of the organoid were used for quantification. The entirety of the organoid section was imaged with a 40X glycerol-immersion objective (2048×2048 -pixel resolution). z stack images were acquired, projected with maximum intensity in Fiji, and deep learning-based segmentation was used to quantify the total number of nuclei and marker-positive nuclei, as aforementioned. The number of marker-positive nuclei was normalized to the total number DAPI⁺ cells.

Dlx1/2b:eGFP area coverage in hTAs: hTAs labeled with LV-Dlx1/2b:eGFP were optically cleared using the CUBIC method, as aforementioned. hTAs were imaged (~ 700 –

1000 μm volume) with a 10X air objective (2048 $\times$ 2048-pixel resolution), and z stack images were used to create a 3D reconstruction of the hCO in Imaris software (Oxford Instruments, v.10.2.0). The nuclear stain signal was used to manually delineate the hCO from the hMGEO and boundary domain (every 10–20 frames), and these regions were then used to create a “surface” to define the hCO part of the hTA. An exemplary output of this workflow is depicted in Figure S11A. The segmented 3D hCO was then projected with maximum intensity in Fiji for analysis. To estimate the percentage of area coverage of Dlx1/2b:eGFP signal in hCOs, images were thresholded (the minimum and maximum values were set similarly between genotypes), and the “Area fraction” was computed.

Cortical interneuron migration: Raw interneuron migration time-lapse imaging files were projected with maximum intensity in Fiji for analysis. Field-of-view drifts occurring during the imaging session were corrected using the “Linear Stack Alignment with SIFT” registration plugin in Fiji. The soma positions of individual interneurons, showing at least two nucleokinesis events throughout the entire imaging session, were outlined using the “Freehand” tool for every frame. To maximize the sample size, only cells that remained within the imaging field for at least 12h were included in the analysis. The centroid coordinates of each outlined soma were extracted as an estimation of its spatial position and were used for downstream analysis. The initial soma position, prior to nucleokinesis, was defined as the centroid of the soma during the onset of soma swelling. The final soma position was defined as the centroid after nucleokinesis was completed. Nucleokinesis length was calculated as the average Euclidean distance between the initial and final soma positions, while nucleokinesis frequency was determined by the total number of nucleokinesis events occurring within a given time period (at least 12h). The total path length was calculated as the sum of Euclidean distances between consecutive soma positions across all frames. Speed was then computed as the total path length divided by the elapsed time required to cover that distance. The path directness ratio was calculated by dividing the total displacement (the straight-line Euclidean distance between the starting and ending soma positions) by the total path length. To measure path complexity, the ratio between the total path length and the diagonal of the minimum bounding rectangle (MBR) of the path was used. Finally, the average turning angle was calculated as the angle between two consecutive movement vectors of the soma. These metrics were computed using R v4.4.2 with a custom script, and the MBR was estimated using the terra v.1.8–5 package.

Calcium imaging recordings: Raw calcium imaging files were processed in Fiji, and the “Mean gray value” was collected for each GCaMP6s⁺ neuron/region of interest (ROI). A representative region of background fluorescence was also collected per field-of-view. Calcium traces were analyzed in R v4.4.2 using a custom script. Background fluorescence was subtracted from each ROI and mean gray values were then transformed to relative changes in fluorescence: $\Delta F(t)/F_0 = (F(t) - F_0)/F_0$, with F_0 being the baseline’s fluorescence, calculated as the average mean gray value across each ROI’s time series. Peak estimation was done with pracma v2.4.4 using the “findpeaks” function (minpeakheight = 0.5, meanpeakdistance = 15). Calcium trace metrics were calculated for each ROI, including inter-event interval (IEI), burst duration, amplitude, and area under the curve. IEI was measured as the average time between consecutive peaks and was estimated for ROIs

exhibiting at least two events during the recording period. Burst duration was defined as the time duration between the onset and offset of each detected event. Amplitude represents the average change in fluorescence from baseline to peak. The average peak's AUC was computed as the integral of the fluorescence signal over time using the “trapz” function of the *pracma* package. Finally, to assess network-related dynamics of calcium signals across fields-of-view, variability in event timing was quantified using the coefficient of variation of IEs for ROIs within each field-of-view, and synchronicity was evaluated as the mean Pearson's correlation coefficient between the fluorescence traces of each ROI pair.

Analysis of MeCP2 expression during fetal development: RNA-sequencing expression data for *MeCP2* was obtained from the BrainSpan project⁵⁵ (<https://www.brainspan.org/static/download.html>). To examine *MeCP2* expression across development, expression values were extracted and merged with sample metadata to associate expression levels with specific brain regions and ages. “All cortical regions” included the dorsolateral prefrontal cortex, ventrolateral prefrontal cortex, anterior (rostral) cingulate (medial prefrontal) cortex, orbital frontal cortex, primary motor-sensory cortex, parietal neocortex, primary somatosensory cortex, posteroventral (inferior) parietal cortex, primary auditory cortex, temporal neocortex, posterior (caudal) superior temporal cortex (area 22c), inferolateral temporal cortex (area 20), occipital neocortex, and primary visual cortex. All data processing and visualization were performed using R v4.4.2, with gene expression trends plotted across developmental time to assess regional and temporal dynamics of *MeCP2* expression.

Single-cell RNA sequencing data for *MeCP2* expression was obtained from cell atlases of the developing neocortex at GW 17–19⁵⁸ and ganglionic eminences at GW 9–12.¹¹⁹ Data pre-processing and dimensionality reduction were performed in R v4.4.2 with Seurat v5.0.0, following the analytical strategies outlined in the original publications. Clustering annotations provided by the respective authors were retained for consistency in downstream analysis. To visualize *MeCP2* expression across cell types, *t*-SNE was used for dimensionality reduction, and expression levels were overlaid onto the *t*-SNE plot.

RNA sequencing

RNA extraction: Total RNA was extracted from pooled organoids, as described above.

Library preparation and sequencing: Bulk RNA-seq libraries were prepared using the Illumina Stranded Total RNA kit (Illumina, 20040529) and pair-end sequenced on a single lane of a NovaSeq X Plus (Illumina) flow cell with a 10% PhiX target.

RNA sequencing data analysis

Data pre-processing: Raw sequencing reads were subjected to quality control and preprocessing using standard pipelines. Initial quality assessment of the raw FASTQ files was performed using FastQC v0.11.9 to evaluate read quality scores, sequence length distribution, and the presence of adapter sequences. Adaptor trimming was carried out using Cutadapt v3.4, specifically targeting Nextera transposase adapter sequences as recommended by Illumina. Post-trimming quality checks were repeated with FastQC to

ensure adapter removal and assess read length distribution after trimming. Trimmed reads were aligned to the human reference genome GRCh38.p14 using Bowtie2 v2.5.1. SAM files generated during alignment were converted to BAM format using SAMtools v1.17. Quantification of gene-level expression was performed using “featureCounts” from the Subread package v2.1.0. to assign reads to genomic features based on the GRCh38.p14 GTF annotation file. rRNA genes were excluded, and low-expressed genes (fewer than 10 counts in over 50% of samples) were filtered out. The final count matrix was used for downstream analyses in R v4.4.2.

Differential gene expression analysis: Differentially gene expression analysis on raw counts was performed using DESeq2 v1.46.0. The design formula used was ~ Differentiation batch + Condition, where “Condition” referred to the genotype of the cell line (2042-MUT vs. 2042-WT). Genes were considered differentially expressed if the adjusted *p*-value (FDR) was below 0.05 and the absolute log2 fold change exceeded 1. Gene Ontology enrichment analysis was performed in ClusterProfiler v4.14.4 using the upregulated genes in MUT-MeCP2. A gene ratio above 5% was used to threshold statistically significant terms. All plots were generated using ggplot2 v3.5.1.

Single-cell RNA sequencing

Generation of single-cell suspension of organoid cells: Individual week 17 old organoids (± 4 days) were harvested, washed once with PBS with 0.1% v/v BSA, and incubated with Accutase/papain mixture (4:1) at 37°C with 5% CO₂. Papain mixture contains papain with L-cysteine and EDTA (Worthington, LK003176) and DNaseI (Worthington, LK003170), both prepared in Earle’s Balanced Salt Solution (EBSS; Worthington, LK003188) and mixed at a ratio of 9:1, respectively, according to the manufacturer’s instructions. After 10 min of incubation, samples were gently triturated with a P1000 pipette, and this process was repeated every 10 min until samples were completely dissociated (for a maximum of 40–50 min incubation time). Papain was inactivated with NDM2, cells were filtered with a 70- μ m Flowmi Cell Strainer (Bel-Art, H13680-0070), and then centrifuged at 200 x g for 7 min. The supernatant was aspirated, leaving a margin of around 30 μ L, and cells were resuspended in 500 μ L of ice-cold PBS with 0.3% v/v BSA. Cells were counted with a NucleoCounter NC-3000 (ChemoMetec), and cell viability was above 90% in all dissociated samples.

Cell hashing: Antibody-based cell hashing was performed using 0.5 μ g of TotalSeq-B anti-human antibodies against β 2-microglobulin and CD298 (BioLegend: 394631, 394633, 394635, and 394637), according to the manufacturer’s instructions.

Library preparation and sequencing: To generate single-cell GEMs (Gel Beads in Emulsion), 40,000 live cells of the multiplexed pool were loaded per channel (to give an estimated recovery of 10,000 cells per channel) onto a Chromium GEM-X Single Cell Chip (10X Genomics, PN-1000690). The gene expression library was prepared with the GEM-X Single Cell 3’ Kit v.4 (10X Genomics, PN-1000691), and the hashtag-oligo library was prepared with the GEM-X Single Cell 3’ Feature Barcode Kit (10X Genomics, PN-1000702). ScRNA-seq libraries were then pooled and pair-end sequenced in a NovaSeq

X flowcell (Illumina) with a targeted sequencing depth per cell of 50,000 gene expression reads and 5,000 surface barcode reads.

Single-cell RNA sequencing data analysis

Data pre-processing: ScRNA-seq reads were processed with Cell Ranger multi v8.0.1 using the prebuilt 10X human GRCh38 reference refdata-gex-GRCh38-2024-A and including introns. Further processing, such as demultiplexing, dimensionality reduction, clustering, and visualization, was performed in R v4.4.2 with Seurat v5.0.0. Demultiplexed datasets were filtered to retain high-quality cells. Only cells containing between 800 and 10,000 sequenced genes, a UMI count below 40,000, and a proportion of mitochondrial and ribosomal genes less than 15% and 30%, respectively, were retained. The Gruffi algorithm¹⁷² (v.1.5.6) was used to identify and remove cells with a transcriptional signature of cellular stress (thresholds: GO:0006096, 1.124; GO:0034976, 0.464; GO:0042063, 0.948). Expression matrices were log-normalized and scaled, and dimensionality reduction was performed using principal component analysis (PCA) on the top 2,000 genes with the highest standardized variance (vst selection method). The top 40 principal components were used to batch correct libraries using Harmony v1.2.3, and the top 25 Harmony embeddings were used for UMAP visualization (for hTAs, the top 18 Harmony embeddings were used).

Cluster annotation: The first 12 harmony embeddings were used to identify telencephalic clusters, with clustering at resolution 0.7 and 1 yielding the most reliable cell type identification. hCO clusters were annotated using known marker genes expressed on the developing human neocortex⁵⁸: aRG (clusters 11 and 12 at resolution 0.7, and cluster 14 at resolution 1), oRG (cluster 10 at resolution 0.7), cycling NPCs (cluster 15 at resolution 0.7, and cluster 17 at resolution 1), IPs (cluster 4 at resolution 0.7, and cluster 2 at resolution 1), excitatory immature neurons (clusters 1, 13, and 14 at resolution 0.7, and clusters 0, 10, and 16 at resolution 1), excitatory infragranular neurons (cluster 3, 6, 8, 9, and 11 at resolution 1), excitatory supragranular neurons (clusters 1, 4, 5, 7, and 15 at resolution 1), and CGE interneurons (cluster 16 at resolution 0.7, and cluster 18 at resolution 1). hTA clusters were annotated similarly, but clustering was only performed at resolution 0.7: RG (cluster 9), cycling NPCs (clusters 10, 15, and 16), IPs (cluster 2), excitatory immature neurons (clusters 0, 1, and 14), excitatory infragranular neurons (cluster 5 and 11), excitatory supragranular neurons (clusters 3, 4, 6, 7, and 8), MGE interneurons (cluster 13), and OPCs (cluster 12).

Comparison of organoid scRNA-seq and fetal atlases: To assess similarity with the developing cerebral cortex and the overall accuracy of supervised cluster annotation, the organoid dataset was compared to a fetal neocortex single-cell dataset retrieved from ref.⁵⁸. The reference dataset was reanalyzed while preserving the original cluster annotations, except for aRG and oRG, which were grouped as “radial glia,” and microglia, pericytes, and endothelial cells, which were grouped as “other” due to their non-ectodermal origin. The top 2000 highly variable features were identified (vst selection method) in both the reference and query datasets, and the average expression was computed for each cluster. Spearman correlation was used to assess transcriptional similarity between the query and reference datasets.

The spatial identity of organoid clusters was evaluated using voxhunt v.1.1.0¹⁷³ by comparing their gene expression profiles to the E15.5 mouse brain reference. *In situ* hybridization micrographs from E15.5 mouse brain were retrieved from the Allen Brain Atlas (ABA) Developing Mouse Brain Data Portal to confirm that the regions of high similarity score of the voxel maps closely align with the spatial localization of the telencephalon and developing cortical plate (<https://developingmouse.brain-map.org/>; *Foxg1*: sagittal, zoomed view of slice 12 of 17; *Eomes*: sagittal, zoomed view of slice 9 of 17; *Satb2*: sagittal, zoomed view of slice 8 of 17). The voxhunt algorithm was further used to assess transcriptional similarity between all cells of the organoid dataset with the brain structures of the BrainSpan Atlas during the second (13–24 GW) and third (25–37 GW) trimester of gestation.

Pseudotime analysis: Pseudotime analysis was carried out using monocle3 v.1.3.7. The Seurat object was converted into a Monocle3 CellDataSet, and its UMAP embeddings were transferred for trajectory inference. Cells were assigned partitions, cluster identities were retained, and trajectory learning was performed based on these embeddings. Finally, cells were ordered along the learned trajectory in pseudotime, and pseudotime values were transferred back to the Seurat object's metadata.

Subsetting of progenitor, neuronal, excitatory, and inhibitory lineage clusters: Dorsal telencephalic progenitors were subsetting from the hCO dataset following a previously published strategy,¹⁷⁴ by selection of: 1) clusters 4, 5, and 6, at resolution 0.2; 2) clusters showing an average expression of *LHX2* and *FOXG1* equal or greater to 1, at resolution 3; 3) pseudotime bin equal or lower than 10 (out of 30 bins). Excitatory neurons were subsetting in a similar way by selection of: 1) clusters 0, 1, 2, 3, and 7, at resolution 0.2; 2) clusters showing an average expression of *NEUROD6* equal or greater to 2 and *FOXG1* equal or greater to 1, at resolution 3; 3) pseudotime bin greater than 10 (out of 30 bins). Cells belonging to the excitatory lineage were subsetting by selection of: 1) clusters 0, 1, 2, 3, 4, and 7 at resolution 0.2; 2) pseudotime bin greater than 3. Excitatory neurons from the hTA dataset were subsetting using the same strategy; however, cluster identities obtained at resolution 0.2 were 0, 1, 2, and 4, and the pseudotime bin cut-off was set at 15. MGE cINs were subsetting by selection of cluster 13 at resolution 0.7.

Differential gene expression analysis: Differentially expressed genes between MUT- and WT-MeCP2 were identified using the Wilcoxon rank-sum test implemented in Presto v.1.0.0. Gene Ontology enrichment analysis was performed in ClusterProfiler v.4.14.4 using the top 300 upregulated or downregulated genes in MUT-MeCP2, for each genetic background. A gene ratio above 5% was used to threshold statistically significant terms.

QUANTIFICATION AND STATISTICAL ANALYSIS

Differentiation batches and/or individual organoids were treated as biological replicates for analysis. Sample sizes were empirically determined based on standards in organoid-related publications and chosen to account for variability between organoids and hPSC lines. Data are presented as mean $\pm$ S.E.M. Statistical analyses were performed using GraphPad Prism v.9.5.0 (525) or R v.4.4.2. Normality was assessed using the Shapiro-Wilk and/or

Kolmogorov-Smirnov tests, and data were considered normally distributed if neither test indicated significance (when applicable, based on sample size). For parametric tests, S.D. was evaluated to ensure homogeneity of variance. The test statistics and significance values for all analyses are reported in the Figure Legends.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Highlights

- RTT cerebral organoids display normal neocortical fate despite *MeCP2* deficiency
- *MeCP2* loss broadly disrupts transcription linked to excitatory neuron maturation
- *MeCP2*-deficient cortical interneurons show hypermotile tangential migration
- Developing cortical circuits need MeCP2 to restrain excitability and synchrony

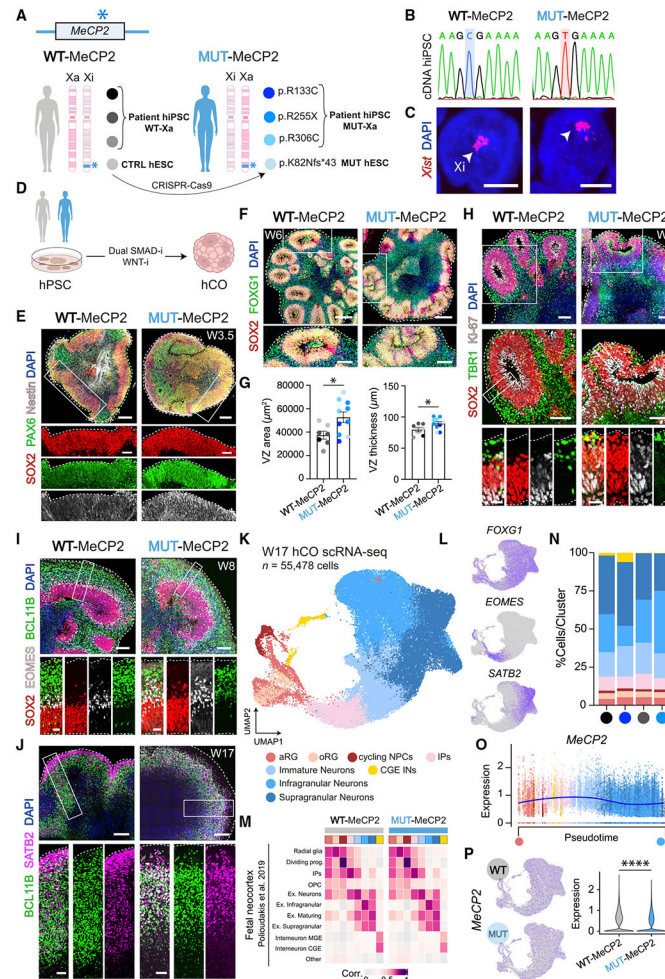


Figure 1. MeCP2 deficiency does not impact a normal acquisition of neocortical fate

(A) Schematic showing the derivation of RTT hiPSC clones based on X chromosome inactivation (Xi) status. In WT-MeCP2 clones, the active X chromosome (Xa) carries a normal copy of *MeCP2*, whereas in MUT-MeCP2 clones, the Xa carries a disease-causing variant of *MeCP2*. An additional hPSC line was used, in which a homozygous deletion was introduced in the *MeCP2* locus via CRISPR-Cas9.

(B) cDNA sequencing of RTT hiPSC lines showing monoallelic expression at the *MeCP2* locus (sequencing traces are from patient 2042).

(C) RNA-FISH for *Xist* in RTT hiPSC lines. Scale bar: 5 μ m.

(D) Schematic detailing the derivation of hCOs from RTT hPSCs.

(E) Week 3.5 RTT hCOs immunostained for progenitor (SOX2/Nestin) and dorsal telencephalic (PAX6) markers. Scale bar: 100 μ m; zoom: 50 μ m.

(F) Week 6 RTT hCOs immunostained for progenitor (SOX2) and anterior forebrain (FOXG1) markers. Scale bar: 200 μ m; zoom: 100 μ m.

(G) Quantification of ventricular zone (VZ) size and thickness in week 5–6 RTT hCOs (WT, $n = 7$, and MUT, $n = 10$ differentiation batches, from 4 hPSC lines, and 1–4 batches and 32–116 VZs analyzed per batch per hPSC line; VZ area: unpaired t test, $t_{(15)} = 2.462$, $*p = 0.0264$; VZ thickness: unpaired t test, $t_{(15)} = 2.163$, $*p = 0.0471$).

- (H) Week 6 RTT hCOs immunostained for apically dividing progenitors (SOX2/Ki67) and pre-plate/layer VI infragranular neuronal (TBR1) markers. Scale bar: 100 μm ; zoom: 50 μm ; 25 μm .
- (I) Week 8 RTT hCOs immunostained for progenitor (SOX2), intermediate progenitor (EOMES), and infragranular/layer V neuronal (BCL11B) markers. Scale bar: 100 μm ; zoom: 25 μm .
- (J) Week 17 RTT hCOs immunostained for infragranular/layer V (BCL11B) and supragranular/layer II–III (SATB2) neuronal markers. Scale bar: 200 μm ; zoom: 50 μm .
- (K) UMAP visualization of single-cell RNA expression of week 17 RTT hCO and unbiased clustering of 55,478 cells showing eight clusters of telencephalic populations: apical radial glia (aRG), outer radial glia (oRG), cycling neural progenitor cells (NPCs), intermediate progenitors (IPs), immature neurons, interneurons with caudal ganglionic eminence identity (CGE INs), infragranular-enriched neurons, and supragranular-enriched neurons.
- (L) *FOXP1*, *EOMES*, and *SATB2* gene expression for all cells.
- (M) Correlation matrix comparing clusters annotated in WT- and MUT-MeCP2 hCOs and cell types in the human fetal cortex atlas from Polioudakis et al.⁵⁸
- (N) Percentage of cells per cluster in each patient.
- (O) Expression of *MeCP2* along pseudotime.
- (P) Expression of *MeCP2* in all WT and MUT single cells (Wilcoxon rank-sum test, **** $p < 0.0001$).

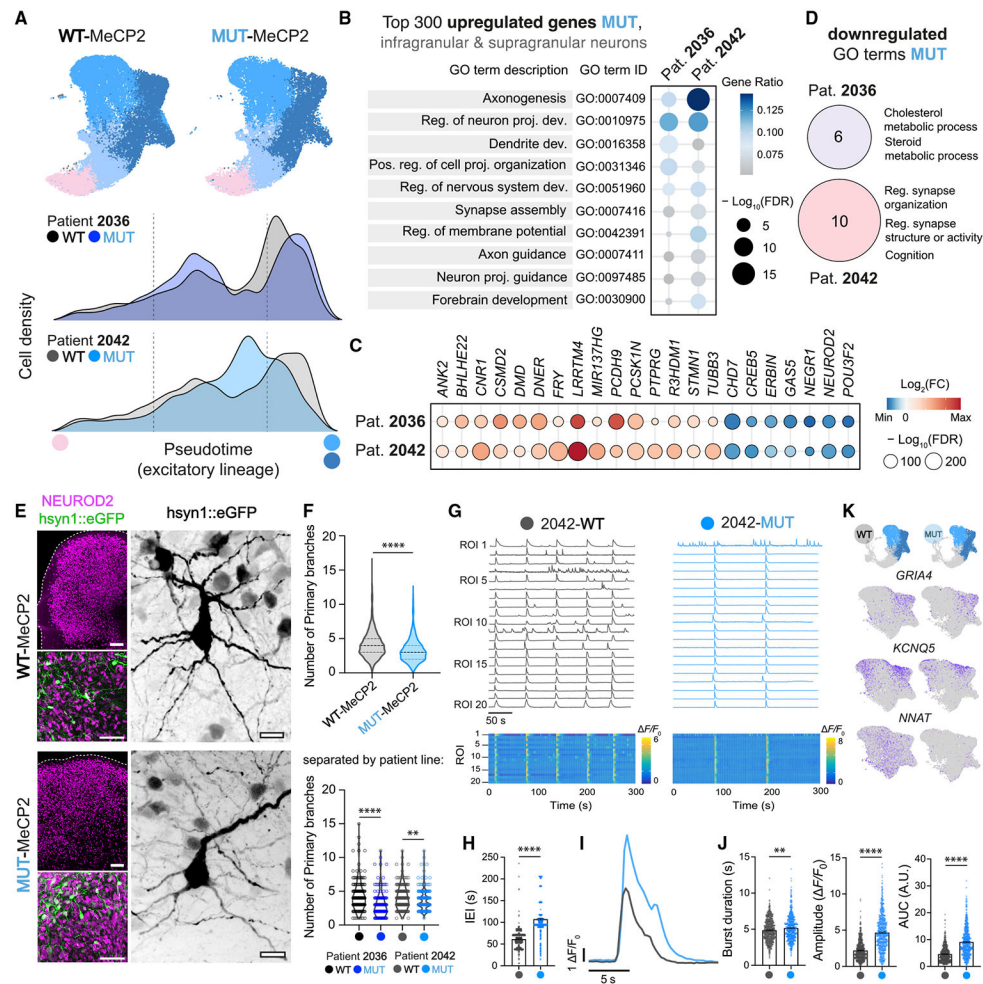


Figure 2. MeCP2 regulates early stages of cortical excitatory neuron maturation

(A) UMAP projection of clusters belonging to the excitatory lineage separated by genotype and pseudotime analysis showing cell densities along the excitatory lineage for each patient.

(B) Gene Ontology (GO) term enrichment analysis of biological processes for the top 300 upregulated genes in MUT-MeCP2 infragranular and supragranular neurons. The top 10 most statistically significant GO terms shared by patients 2036 and 2042 are shown (gene ratio above 5% and ordered by FDR).

(C) Selected differentially expressed genes in MUT neurons showing similar trends in both patients.

(D) GO term enrichment analysis of biological processes for the top 300 downregulated genes in MUT-MeCP2 infragranular and supragranular neurons. No overlapping GO terms were found between patients 2036 and 2042 (gene ratio above 5% and filtered by FDR).

(E) Left, week 17 RTT hCOs immunostained for the excitatory neuronal marker NEUROD2, showing expression in the cortical plate domain. Scale bar: 100 μ m. Neurons infected with AAV7m8-hsyn1::eGFP show co-expression of eGFP with NEUROD2. Scale bar: 50 μ m. Right, representative micrographs of week 17 eGFP-labeled RTT cortical neurons. Scale bar: 10 μ m.

(F) Quantification of the number of primary branches of week 17 eGFP-labeled RTT cortical neurons (WT, $n = 412$ cells, from 2 hiPSC lines, 3 organoids per line, and 3–4 fields-of-view analyzed per organoid: 2036-WT, $n = 188$, and 2042-WT, $n = 224$ cells; MUT, $n = 407$ cells, from 2 hiPSC lines, 3 organoids per line, and 3–4 fields-of-view analyzed per organoid: 2036-MUT, $n = 187$, and 2042-MUT, $n = 220$ cells. WT vs. MUT, Mann-Whitney test, $U = 59971$, **** $p < 0.0001$; 2036, Mann-Whitney test, $U = 9932$, **** $p < 0.0001$; 2042, Mann-Whitney test, $U = 20997$, ** $p = 0.0057$).

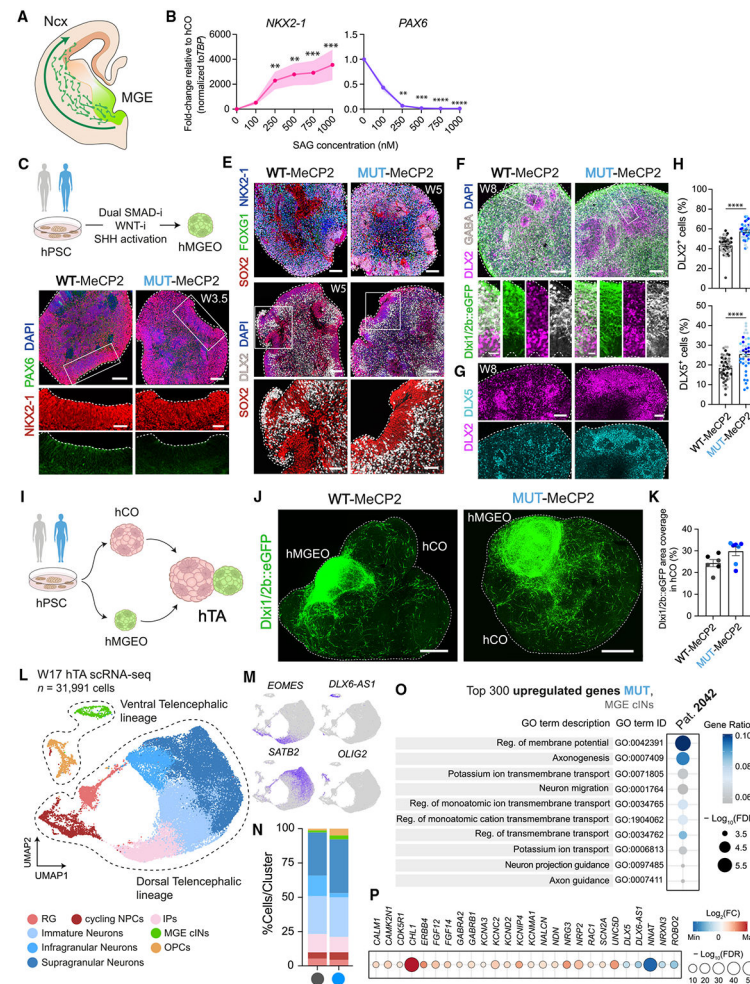
(G) Representative calcium traces and heatmap showing $\Delta F/F_0$ of GCaMP6s from 20 ROIs of RTT hCO neurons derived from patient 2042 at week 17 of differentiation.

(H) Quantification of average inter-interval event (IEI) duration in RTT hCO neurons (2042-WT, $n = 661$ ROIs, and 2042-MUT, $n = 603$ ROIs, from 5 organoids and 3–4 fields-of-view per hiPSC line; Mann-Whitney test, $U = 27735$, **** $p < 0.0001$).

(I) Representative calcium peaks from RTT hCO neurons derived from patient 2042 at week 17 of differentiation.

(J) Quantification of average burst duration, amplitude, and area under the curve (AUC) in RTT hCO neurons (2042-WT, $n = 665$ ROIs, and 2042-MUT, $n = 632$ ROIs, from 5 organoids per hiPSC line and 3–4 fields-of-view per organoid; burst duration: Mann-Whitney test, $U = 192504$, ** $p = 0.0089$; amplitude: Mann-Whitney test, $U = 78215$, **** $p < 0.0001$; AUC: Mann-Whitney test, $U = 91394$, **** $p < 0.0001$).

(K) Expression, separated by genotype, of *GRIA4*, *KCNQ5*, and *NNAT* in infragranular and supragranular RTT hCO neurons at week 17 of differentiation.



(F) Week 8 hMGEOs derived from *Dlxi1/2b::eGFP* reporter hPSC lines immunostained for GFP alongside cINP/cortical interneuron (cIN; DLX2) and inhibitory neuronal markers. Scale bar: 100 μm ; zoom: 25 μm .

(G) Week 8 hMGEOs immunostained for the IP/cIN marker DLX2 and the cIN marker DLX5. Scale bar: 100 μm .

(H) Quantification of the percentage of DLX2⁺ and DLX5⁺ nuclei in RTT hMGEOs in all cell lines at week 8 of differentiation (WT-MeCP2, $n = 49$ and MUT-MeCP2, $n = 39$ organoids, from 4 hPSC lines per condition and 1–3 differentiation batches per hPSC line; DLX2: Mann-Whitney test, $U = 178$, **** $p < 0.0001$; DLX5: unpaired t test with Welch's correction, $t_{(65.78)} = 4.406$, **** $p < 0.0001$).

(I) Schematic illustrating the generation of hTAs from RTT hPSCs.

(J) Representative images of optically cleared RTT hTAs at 4 weeks post-fusion. The hMGEO was labeled with LV-*Dlxi1/2b::eGFP*. Scale bar: 500 μm .

(K) Quantification of area coverage percentage of *Dlxi1/2b::eGFP*⁺ signal in segmented hCOs at 4 weeks post-fusion (WT, $n = 6$, and MUT, $n = 7$ assembloids from 2 hiPSC lines per condition; Mann-Whitney test, $U = 8$, $p = 0.0734$).

(L) UMAP visualization of single-cell RNA expression of week 17 RTT hTAs derived from patient 2042 and unbiased clustering of 31,991 cells showing eight clusters of telencephalic populations: radial glia (RG), cycling neural progenitor cells (NPCs), intermediate progenitors (IPs), immature neurons, infragranular-enriched neurons, supragranular-enriched neurons, cortical interneurons (INs) with medial ganglionic eminence (MGE) identity, and oligodendrocyte progenitor cells (OPCs).

(M) *EOMES*, *SATB2*, *DLX6-AS1*, and *OLIG2* gene expression for all cells.

(N) Percentage of cells per cluster per genotype for patient 2042.

(O) Gene Ontology (GO) term enrichment analysis of biological processes for the top 300 upregulated genes in MUT-MeCP2 MGE cINs derived from patient 2042. The top 10 most statistically significant GO terms are shown (gene ratio above 5% and ordered by FDR).

(P) Selected differentially expressed genes in MUT MGE cINs for patient 2042.

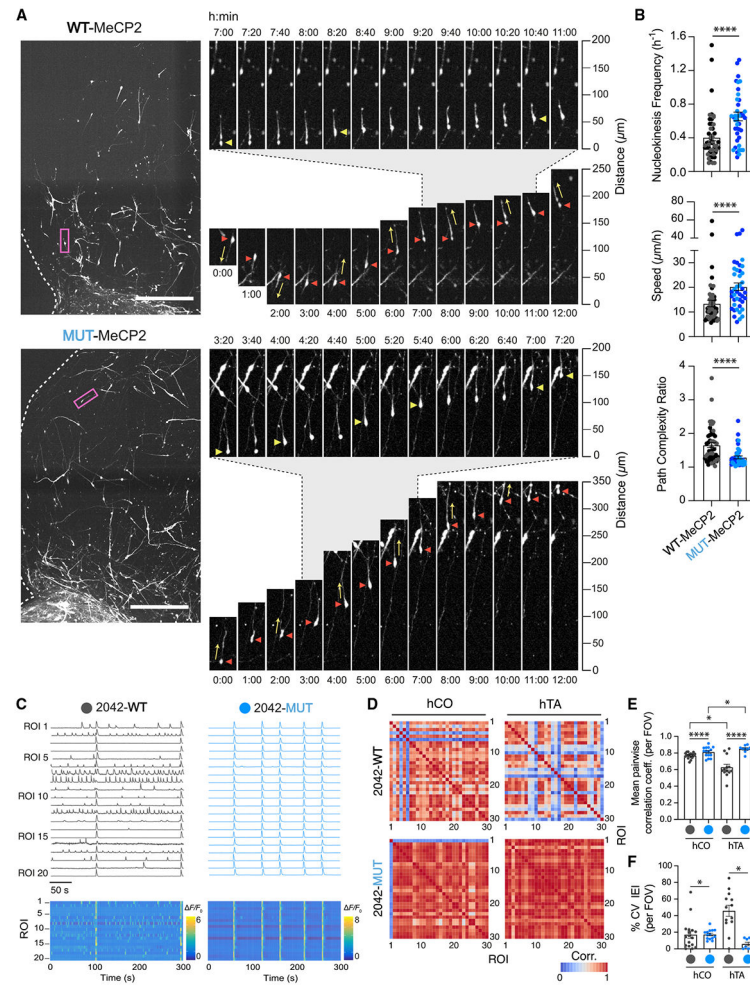


Figure 4. MeCP2 deficiency promotes cortical interneuron migration and increases the susceptibility of cortical excitatory-inhibitory neuronal circuits to hyperexcitability and hypersynchrony

(A) Representative time-lapse images of cINs undergoing tangential migration within the hCO region of the hTA at 4 weeks post-fusion. Scale bar: 300 μm . For each genotype, the right panels show the selected *Dlx1/2b::eGFP⁺* cIN within the wider hTA context. In the lower panels, the migratory trajectory of the highlighted cell is shown over a 12-h period; red dots mark the soma position at each time point, and yellow arrows indicate the direction of movement. Upper panels display a representative 4-h segment capturing nucleokinesis events; yellow markers denote the new soma position following each nucleokinesis.

(B) Quantification of nucleokinesis frequency, speed of movement (over the entire course), and path complexity ratio of RTT cINs (WT, $n = 49$ cells, from 2 hiPSC lines and 3 assembloids per line: 2036-WT, $n = 28$, and 2042-WT, $n = 21$ cells; MUT, $n = 42$ cells, from 2 hiPSC lines and 3 assembloids per line: 2036-MUT, $n = 23$, and 2042-MUT, $n = 19$ cells. Nucleokinesis frequency: Mann-Whitney test, $U = 536$, **** $p < 0.0001$; speed: Mann-Whitney test, $U = 503$, **** $p < 0.0001$; path complexity ratio: Mann-Whitney test, $U = 529$, **** $p < 0.0001$).

(C) Representative calcium traces and heatmaps showing $\Delta F/F_0$ of GCaMP6s from 20 ROIs of RTT hTA neurons derived from patient 2042 at week 17 of differentiation (approximately 2 months post-fusion).

(D) Correlation matrices showing correlated activity across ROIs in representative fields for hCOs and hTAs derived from patient 2042.

(E) Quantification of mean pairwise correlation coefficient and (F) inter-event interval coefficient of variance (CV IEI; in percentage) per field-of-view in hCOs and hTAs from patient 2042 (hCO, $n = 14\text{--}17$ fields from 5 organoids per hiPSC line; hTA, $n = 9\text{--}13$ fields from 3 to 4 assembloids per hiPSC line. Mean pairwise correlation coeff.: ordinary two-way ANOVA, $F_{(1,51)} = 7.272$ (for hCO vs. hTA, $p = 0.0095$), $F_{(1,51)} = 23.33$ (WT vs. MUT, $p < 0.0001$), $*p < 0.05$ and $****p < 0.0001$ with Holm-Šidák's multiple comparison test. CV IEI: ordinary two-way ANOVA, $F_{(1,50)} = 4.826$ (for hCO vs. hTA, $p = 0.0327$), $F_{(1,50)} = 9.646$ (WT vs. MUT, $p = 0.0031$), $*p < 0.05$ with Holm-Šidák's multiple comparison test.

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
BCL11B (CTIP2)	abcam	Cat# ab18465; RRID:AB_2064130
Doublecortin (DCX)	aves	Cat# DCX; RRID:AB_2313540
DLX2	Santa Cruz	Cat# sc-393879; RRID:AB_3678657
DLX5	R&D Systems	Cat# AF6710; RRID:AB_10890008
EOMES (TBR2)	R&D Systems	Cat# AF6166; RRID:AB_10569705
FOXP1	abcam	Cat# ab196868; RRID:AB_2892604
GABA	Sigma-Aldrich	Cat# A2052; RRID:AB_477652
GAD67 (GAD1)	EMD Millipore	Cat# MAB5406; RRID:AB_2278725
GSH2 (GSX2)	EMD Millipore	Cat# ABN162; RRID:AB_11214376
GFP	GeneTex	Cat# GTX13970; RRID:AB_371416
Ki-67	BD Transduction	Cat# 550609; RRID:AB_393778
LHX2	abcam	Cat# ab184337; RRID:AB_2916270
MeCP2	Cell Signaling	Cat# 3456S; RRID:AB_2143849
MAP2	Invitrogen	Cat# PA1-10005; RRID:AB_1076848
NANOG	R&D Systems	Cat# AF1997; RRID:AB_355097
Nestin	abcam	Cat# ab22035; RRID:AB_446723
NEUROD2	abcam	Cat# ab104430; RRID:AB_10975628
NKX2-1 (TTF-1)	Leica/Novocastra	Cat# NCL-L-TTF-1; RRID:AB_564042
OCT4 (POU5F1)	Millipore	Cat# MAB4419; RRID:AB_1977399
PAX6	BioLegend	Cat# 901301; RRID:AB_2565003
PKC α	BD Transduction	Cat# 610207; RRID:AB_397606
SATB2	abcam	Cat# ab51502; RRID:AB_882455
SOX2	R&D Systems	Cat# AF2018; RRID:AB_355110
SOX2	abcam	Cat# ab97959; RRID:AB_2341193
TBR1	abcam	Cat# ab31940; RRID:AB_2200219
TBR1	abcam	Cat# ab183032; RRID:AB_2936859
Goat anti-Chicken IgG (H + L) Alexa Fluor™ 555	Invitrogen	Cat# A-21437; RRID:AB_1500593
Donkey anti-Mouse IgG (H + L) Alexa Fluor™ 647	Invitrogen	Cat# A-31571; RRID:AB_162542
Donkey anti-Rabbit IgG (H + L) Alexa Fluor™ 488	Invitrogen	Cat# A-21206; RRID:AB_2535792
Donkey anti-Rabbit IgG (H + L) Alexa Fluor™ 555	Invitrogen	Cat# A-31572; RRID:AB_162543
Donkey anti-Rabbit IgG (H + L) Alexa Fluor™ 647	Invitrogen	Cat# A-31573; RRID:AB_2536183
Donkey anti-Goat IgG (H + L) Alexa Fluor™ 488	Invitrogen	Cat# A-11055; RRID:AB_2534102
Donkey anti-Goat IgG (H + L) Alexa Fluor™ 555	Invitrogen	Cat# A-21432; RRID:AB_141788
Donkey anti-Goat IgG (H + L) Alexa Fluor™ 647	Invitrogen	Cat# A-21447; RRID:AB_141844
Donkey anti-Rat IgG (H + L) Alexa Fluor™ 488	Invitrogen	Cat# A-21208; RRID:AB_141709
Donkey anti-Sheep IgG (H + L) Alexa Fluor™ 488	Invitrogen	Cat# A-11015; RRID:AB_141362
Donkey anti-Sheep IgG (H + L) Alexa Fluor™ 555	Invitrogen	Cat# A-21436; RRID:AB_2535857
Donkey anti-Chicken IgY(IgG) (H + L) Alexa Fluor® 488-conjugated AffiniPure	Jackson ImmunoResearch	Cat# 703-545-155; RRID:AB_2340375

REAGENT or RESOURCE	SOURCE	IDENTIFIER
TotalSeq™-B0251 anti-human Hashtag 1	BioLegend	Cat# 394631; RRID:AB_2814347
TotalSeq™-B0252 anti-human Hashtag 2	BioLegend	Cat# 394633; RRID:AB_2814348
TotalSeq™-B0253 anti-human Hashtag 3	BioLegend	Cat# 394635; RRID:AB_2814349
TotalSeq™-B0254 anti-human Hashtag 4	BioLegend	Cat# 394637; RRID:AB_2814350
Chemicals, peptides, and recombinant proteins		
2-methylbutane	Sigma-Aldrich	Cat# MX0760
β -Mercaptoethanol	Life Technologies	Cat# 31350-010
Accutase	Life Technologies	Cat# A11105-01
Antibiotic-antimycotic (100X)	Life Technologies	Cat# 15240062
B-27® supplement (50X)	Life Technologies	Cat# 17504-001
B-27® supplement without vitamin A (50X)	Life Technologies	Cat# 12587-001
BDNF	PeprTech	Cat# 450-02
bFGF/FGF2	Life Technologies	Cat# PHG0261
BrainPhys™ imaging optimized medium	StemCell Technologies	Cat# 05796
BrainPhys™ neuronal medium	StemCell Technologies	Cat# 05790
Chemically Defined lipid concentrate	Life Technologies	Cat# 11905031
DAPI	Sigma-Aldrich	Cat#D9564
Dibutyl cAMP	Sigma-Aldrich	Cat#D0627
DMEM-F12 with HEPES	Life Technologies	Cat# 11330-032
DNaseI	Worthington	Cat# LK003170
DNaseI recombinant, RNase-free	Roche	Cat# 04716728001
DpnI	NEB	Cat# R0176S
Earle's Balanced Salt Solution	Worthington	Cat# LK003188
EGF	PeprTech	Cat# 100-15
Formaldehyde 16% (w/v), methanol-free	ThermoFisher Scientific	Cat# 28908
Formamide	Sigma-Aldrich	Cat# S4117
GDNF	PeprTech	Cat# 450-10
Gelatin from porcine skin	Sigma-Aldrich	Cat# G1890
Glucose solution	Life Technologies	Cat# A2494001
GlutaMAX™ supplement	Life Technologies	Cat# 35050-061
Insulin solution human	Sigma-Aldrich	Cat# 19278
Knock-Out Serum Replacement	Life Technologies	Cat# 10828-028
L-ascorbic acid	Sigma-Aldrich	Cat# A4403
LDN-193189	Sigma-Aldrich	Cat# SML0559
Matrigel®	Corning	Cat# 354234
Matrigel® hESC-Qualified Matrix	Corning	Cat# 354277
MEM-NEAA (100X)	Life Technologies	Cat# 11140-050
mTeSR™ Plus	StemCell Technologies	Cat# 100-0276
N2 supplement (100X)	Life Technologies	Cat# 17502-001
Neurobasal™	Life Technologies	Cat# 21103-049
Papain with L-cysteine and EDTA	Worthington	Cat# LK003176

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Penicillin-streptomycin (10,000 U/mL)	Life Technologies	Cat# 15140122
Phenol:chloroform:isoamyl alcohol	Sigma-Aldrich	Cat# P2069-100ML
PrimeSTAR® GXL DNA high-fidelity polymerase	Takara	Cat# R050A
Proteinase K	Invitrogen	Cat# EO491
ProLong™ Gold antifade reagent	Invitrogen	Cat# P36930
Puromycin	InvivoGen	Cat# ant-pr-1
RedDot™2 Far-Red Nuclear Stain	biotium	Cat# 40061
ReLeSR™ Reagent	StemCell Technologies	Cat# 100-0484
SAG	Selleckchem	Cat# S7779
SB-431542	Stemgent	Cat# 04-0010
Stellaris® FISH Probes, Human <i>XIST</i> with Quasar® 570 Dye	Biosearch Technologies	Cat# SMF-2038-1
Sucrose	Sigma-Aldrich	Cat# S0389
Tissue-Clearing Reagent CUBIC-L	TCI	Cat# T3740
Tissue-Clearing Reagent CUBIC-R+(M)	TCI	Cat# T3741
TRIzol™ Reagent	Invitrogen	Cat# 15596026-100ML
XAV-939	Sigma-Aldrich	Cat# X3004
Xpert Fast Hotstart Mastermix (2X)	grisp	Cat# GE35.5001
Y-27632 (ROCK inhibitor)	Tocris	Cat# 1254
Critical commercial assays		
CytoTune™-iPS 2.0 Sendai Reprogramming Kit	Invitrogen	Cat# A16518
Illumina Stranded Total RNA Kit	Illumina	Cat# 20040529
GEM-X Single Cell 3' Kit v.4	10X Genomics	Cat# PN-1000691
GEM-X Single Cell 3' Feature Barcode Kit	10X Genomics	Cat# PN-1000702
Monarch® PCR & DNA cleanup Kit	NEB	Cat# T1030S
NEBuilder® HiFi DNA Assembly Cloning Kit	NEB	Cat# E5520S
QuantiTec Reverse Transcription Kit	QIAGEN	Cat# 205313
RT ² SYBR® Green Fluor qPCR Mastermix	QIAGEN	Cat# 330523
Stellaris RNA-FISH hybridization buffer	Biosearch Technologies	Cat# SMF-HB1-10
Stellaris RNA-FISH Wash Buffer A	Biosearch Technologies	Cat# SMF-WA1-60
Stellaris RNA-FISH Wash Buffer B	Biosearch Technologies	Cat# SMF-WB1-20
Deposited data		
Processed bulk RNA-seq data	This study	GEO: GSE297594
Processed single-cell RNA-seq data	This study	GEO: GSE297594
Experimental models: Cell lines		
Human: hESC, WA09 (H9)	WiCell	RRID:CVCL_9773
Human: hiPSC, KOLF2.1J (WTSli018-B-12)	The Jackson Laborator ¹⁶⁶	RRID:CVCL_B5P3
Human: hiPSC, WT83.06	Trujillo et al. ¹⁶⁷	N/A
Human: hiPSC, WT.Fc02	This study	N/A
Human: RTT hiPSC, 2036-WT (p.R133C-WT)	RSRT iPSC Collection	RS iPSC 2036-0033A
Human: RTT hiPSC, 2036-MUT (p.R133C-MUT)	RSRT iPSC Collection	RS iPSC 2036-0036C
Human: RTT hiPSC, 2042-WT (p.R255X-WT)	RSRT iPSC Collection	RS iPSC 2042-0047B

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Human: RTT hiPSC, 2042-MUT (p.R255X-MUT)	RSRT iPSC Collection	RS iPSC 2042-0035E
Human: RTT hiPSC, 2047-WT (p.R306C-WT)	RSRT iPSC Collection	RS iPSC 2047-0039E
Human: RTT hiPSC, 2047-MUT (p.R306C-MUT)	RSRT iPSC Collection	RS iPSC 2047-0022G
Human: <i>MeCP2</i> ^{+/+} hESC, 2982-WT (p.K82Nfs*43-WT)	Trujillo et al. ⁶⁸	N/A
Human: <i>MeCP2</i> ^{-/-} hESC, 2982-MUT (p.K82Nfs*43-MUT)	Trujillo et al. ⁶⁸	N/A
Oligonucleotides		
Primer sequences used for genotyping (Table S2)	This study	N/A
Primer sequences used for allelic expression (Table S3)	This study	N/A
Primer sequences used for RT-qPCR (Table S4)	This study	N/A
Recombinant DNA and viral vectors		
pLenti-CMV::GFP-WPRE-Puro	addgene	Cat# 17448
pLenti-Dlxi1/2b:eGFP	Gift from Sergiu P. Paşca and John L. Rubenstein	N/A
pLenti-Dlxi1/2b:eGFP-WPRE-Puro	This study	N/A
LV-Dlxi1/2b:eGFP	VectorBuilder	N/A
LV-Dlxi1/2b:eGFP-WPRE-Puro	VectorBuilder	N/A
AAV7m8-hsyn1-eGFP	Unitat de Producció de Vectors, Universitat Autònoma de Barcelona	N/A
pAAV-hsyn1-GCaMP6s-P2A-nls-dTomato	addgene	Cat# 17448
AAV7m8-hsyn1-GCaMP6s-P2A-nls-dTomato	VectorBuilder	N/A
Software and algorithms		
Adobe Illustrator (CC)	Adobe	https://www.adobe.com/products/illustrator.html
CellRanger	10X Genomics	https://www.10xgenomics.com/support/software/cell-ranger/latest
DESeq2	Bioconductor	https://bioconductor.org/packages/release/bioc/html/DESeq2.html
GraphPad Prism	GraphPad Software Inc., La Jolla, CA, USA	https://www.graphpad.com/features
ImageJ (Fiji)	ImageJ	https://imagej.net/software/fiji/
Imaris	Oxford Instruments	https://imaris.oxinst.com
R	R Core Team	https://www.r-project.org/
RStudio	Posit	https://posit.co/download/rstudio-desktop/
Seurat	Satija Lab	https://satijalab.org/seurat/
Other		
6-well plate for iPSC culture	Corning	Cat# 3516
96-well glass bottom microplate	ibidi	Cat# 89627
Aggrewell™800 plate	StemCell Technologies	Cat# 34815
Fisherbrand™ Superfrost™ Plus microscope slides	FisherScientific	Cat# 12-550-15
Ultra-low attachment 6-well plate	Corning	Cat# 3471